

CS (Diploma) — Hardware, Devices, Storage and the Motherboard

Elective 3: **Computer Science / Computer Engineering (Diploma)**, NPSC CTSE 2026.

Covers **Paper-I units 1, 2, 3 and 4**. Companion to [CSDip_00_OVERVIEW.md](#).

Drill questions for this material are in [CSDip_04_MCQ_and_QuestionBank.md](#).

Why this file matters

This is the **most MCQ-dense file in the whole set**, and it is the one where a B.Tech CS graduate is most likely to be quietly under-prepared.

Here is the awkward truth about your position. Your degree taught you *computer architecture* — pipelines, cache coherence, addressing modes, the memory hierarchy as a performance argument. It almost certainly did **not** teach you that the floppy connector has 34 pins, that the PS/2 mouse port is green, that a 3.5-inch HD floppy holds 1.44 MB, or that a CD's laser is 780 nm. Those are exactly the facts a Diploma paper asks, because they are crisp, unambiguous and easy to write four options for.

The real 2024 CS (Diploma) Paper-I confirms this. Around **17% of it** — roughly 35 of 200 questions — sat in the hardware / motherboard / storage / bus block, alongside another 18 questions on I/O devices and 16 on generations, classification and number systems. Together that is **over a third of Paper-I** drawn from this single file's material. The questions were things like: *"Which is a major circuitry that provides support and control for main memory, cache memory and PCI bus controllers?"* (north bridge), *"Which defines a motherboard's size, shape and how it is mounted to the case?"* (form factor), *"If there are continuous short beeps there is a —"* (system board failure), and a matching question on the capacities of CD, DVD, floppy and Blu-ray.

None of that is hard. All of it is **learnable in a fixed number of hours**, which is the best property a topic can have three weeks out. Numbers memorised here convert directly into marks.

The one section to over-invest in: §9, the motherboard. It is simultaneously a guaranteed 15-mark descriptive question, a rich source of MCQs, and the thing you know least well.

How to use this file

1. **Read §1–§5 once** — they are short and mostly familiar.
2. **Drill §4 (number systems) with a pen**, not by reading. Speed matters more than understanding here; you already understand it.
3. **Learn §6–§8 as facts**. Use the tables. Cover the right-hand column and recite.
4. **Hand-draw the §9 motherboard diagram three times** on separate days.
5. Finish with the **"Numbers you must memorise"** table at the end — it is the single highest-yield half-page in this folder.

Topic checklist

Tick only when you can write the answer from memory, not just recognise it.

- [] §1 Generations of computers — five-row table with technology, examples, limitations
- [] §2 Classification — super / mainframe / mini / workstation / micro, with the comparison table
- [] §3 Data representation — bit/byte/nibble/word, BCD, ASCII, EBCDIC, Unicode
- [] §4a Number systems and all six conversion directions

- [] §4b Binary arithmetic — addition, subtraction
- [] §4c 1's complement subtraction with end-around carry
- [] §4d 2's complement subtraction, both signs of result
- [] §5 Functional block diagram — hand-drawable, with data and control paths distinguished
- [] §6 Input devices — keyboard, mouse, scanners, OCR/OMR/MICR, bar-code
- [] §7a Output — CRT vs LCD working and comparison
- [] §7b Printers — the three-way comparison table, and the working of each
- [] §7c Plotters
- [] §8a ROM family — ROM/PROM/EPROM/EEPROM/Flash comparison
- [] §8b Registers, cache, RAM; SRAM vs DRAM; SDRAM vs DDR
- [] §8c Memory hierarchy table, capacity and speed
- [] §8d Floppy disks and their four capacities
- [] §8e Hard disk — components, organisation, capacity and access-time sums
- [] §8f Drive interfaces — IDE/PATA, SATA, SCSI
- [] §8g Optical storage — CD/DVD organisation and read/write mechanism
- [] §9 **Motherboard in full detail — labelled diagram drawn from memory**
- [] Numbers table memorised and self-tested

1. Generations of computers

Concept

A "generation" is not a marketing label — it is defined by **the physical device used to switch electrical signals**. Everything else about the machine follows from that one choice.

Think about why. If your switch is a vacuum tube, it is the size of a light bulb, needs a heated filament, draws watts and burns out after a few thousand hours. A machine with 18,000 of them must be room-sized, must have air conditioning, and will fail somewhere every few minutes. Shrink the switch to a transistor and the same logic fits in a cabinet, runs cool and stays up for weeks. Put thousands of transistors on one chip and it fits on a desk. Put millions on one chip and it fits in a pocket.

So the whole story of computing is one sentence: the switch got smaller. Size, cost, power and failure rate fell; speed, reliability and memory rose. If you remember only that, you can reconstruct most of the table.

Key points — the table to memorise

	1st Generation	2nd Generation	3rd Generation	4th Generation	5th Generation
Period	1946–1958	1959–1964	1965–1971	1971–present	Present / future
Switching technology	Vacuum tube (thermionic valve)	Transistor	Integrated Circuit (SSI, MSI)	LSI / VLSI — the microprocessor	ULSI , parallel processing, AI
Main memory	Magnetic drum, delay lines	Magnetic core	Magnetic core → early semiconductor	Semiconductor RAM (DRAM)	High-density semiconductor
Secondary storage	Punched cards, paper tape	Magnetic tape, magnetic disk	Magnetic disk	Hard disk, floppy, optical, SSD	SSD, cloud

Language	Machine language (binary)	Assembly; early HLL — FORTRAN, COBOL	HLL — FORTRAN IV, COBOL, PASCAL, BASIC	HLL, 4GL, GUI, OOP	Natural language, AI languages
Operating system	None — manual operation	Batch monitor	Multiprogramming, time-sharing	GUI, multitasking, networked	Distributed, intelligent
Speed	Milliseconds	Microseconds	Nanoseconds	Nanoseconds / picoseconds	Faster still
Size / power	Room-sized, tens of kW	Cabinet-sized	Desk-sized	Desktop → pocket	Miniature, embedded
Reliability	Very poor	Better	Good	Very high	Very high
Examples	ENIAC, EDVAC, EDSAC, UNIVAC-I, IBM 701	IBM 1401, IBM 7090/7094, CDC 1604, Honeywell 400	IBM 360/370, PDP-8, PDP-11, ICL 2900	IBM PC, Apple II, Pentium, all modern PCs	Robots, expert systems, quantum research
Chief limitation	Huge, hot, unreliable, hand-wired	Still needed AC; costly	Costly to maintain; needed AC	—	—

Facts most often turned into MCQs

Fact	Answer
Who invented the transistor, and where	Bardeen, Brattain, Shockley — Bell Laboratories, 1947
Who invented the IC	Jack Kilby (TI) and Robert Noyce (Fairchild), 1958–59
First microprocessor	Intel 4004, 1971, 4-bit
Generation using ICs	Third
Generation where GUIs were developed	Fourth
First completely automatic general-purpose programmable digital computer	Mark I (Harvard Mark I, 1944) — ⚠ some books say ENIAC; the 2024 paper offered Mark I, Analytical Engine, Colossus and ENIAC as options. Mark I is the safer answer for "automatic general-purpose programmable"; ENIAC is the answer for "first general-purpose <i>electronic</i> "
Characteristic feature of 1st generation	Bulky (not flexible, not portable, not easy to program)
A 2nd-generation machine among ENIAC / EDSAC / IBM 7090 / UNIVAC	IBM 7090
Father of the computer	Charles Babbage (Analytical Engine, 1837)
Stored-program concept	John von Neumann, 1945

Characteristics of a computer (a stock 5-marker)

Speed · Accuracy · Diligence (no fatigue, no boredom) · Versatility · Storage capacity ·

Automation · Reliability. **A computer has no IQ, no intuition and no common sense** — it cannot decide anything not programmed into it. That last point is a favourite "which is **not** a characteristic" MCQ (2024 P-I Q182 offered Accuracy / IQ / Diligence / Efficiency; the answer is **IQ**).

Accuracy note: a computer's accuracy depends on its configuration and on the instructions given to it, **not** on its physical size. Errors are almost always **GIGO** — garbage in, garbage out.

Likely exam questions

- Explain the generations of computers with the technology, characteristics and examples of each. [10]
- Write short notes on the fifth generation of computers. [5]
- What are the characteristics of a computer? [5]

MCQ traps

- "Which generation used ICs?" — Third. Do not confuse the IC (3rd) with the microprocessor (4th). The microprocessor *is* an IC, but the generation break is at LSI/VLSI.
- **UNIVAC-I is first generation, not second** — the name sounds modern.
- **Transistor ≠ IC**. 2nd gen used discrete transistors, 3rd put many on one chip.

2. Classification of computers

Concept

Two computers can both be "fast" in completely different senses. A supercomputer is fast at doing **one enormous calculation**; a mainframe is fast at doing **an enormous number of small transactions**. That distinction — raw floating-point throughput versus transaction and I/O throughput — is the axis the whole classification turns on, and it is exactly what examiners test.

- **Supercomputer:** "How quickly can I simulate tomorrow's weather?" — measured in **FLOPS**.
- **Mainframe:** "How many bank withdrawals can I process this second, without ever losing one?" — measured in **transactions per second**, with legendary reliability and I/O capacity.
- **Minicomputer:** a department's shared machine — the middle ground, now largely extinct.
- **Workstation:** one user, but a demanding one — CAD, 3-D rendering, simulation.
- **Microcomputer:** one user, ordinary demands — the PC.

Comparison table

Feature	Supercomputer	Mainframe	Minicomputer	Workstation	Microcomputer
Users supported	Few (batch jobs)	Thousands , via terminals	Tens to hundreds	One	One
Processors	Thousands (massively parallel)	Several to hundreds	Few	1–4	1–2
Speed measure	FLOPS (TFLOPS/ PFLOPS)	Transactions/ sec, MIPS	MIPS	MIPS/FLOPS	GHz
Main memory	Terabytes+	Hundreds of GB – TB	GB	GB – tens of GB	GB
Cost	₹ hundreds of crores	₹ crores	₹ lakhs	₹ lakhs	₹ tens of thousands

Likely exam questions

- Classify computers on the basis of size and capacity, and compare them. [10]
- Differentiate between a supercomputer and a mainframe computer. [5]
- Differentiate between analog, digital and hybrid computers. [5]

MCQ traps

- "Fastest computer" = supercomputer; "most users / highest throughput" = mainframe. Read which one is asked.
- "Least powerful" among mini/micro/super/mainframe is the microcomputer.
- A workstation is single-user even though it is more powerful than a PC. It is not a server.

3. Data representation and coding schemes

Concept

A computer has exactly one thing it can physically represent: **the presence or absence of a voltage**. Everything — text, images, sound, this sentence — has to be encoded as a pattern of those two states. A "coding scheme" is simply an agreed dictionary saying which bit pattern means which character. There is nothing deep here; ASCII says 1000001 means A because a committee decided so in 1963.

Units

Unit	Size	Note
Bit	1 binary digit, 0 or 1	Smallest unit of data. From "binary digit"
Nibble	4 bits	Exactly one hexadecimal digit; half a byte
Byte	8 bits	Smallest addressable unit; holds one ASCII character
Word	16 / 32 / 64 bits	The number of bits the CPU handles as one unit — sets register and data-bus width. Word length is machine-dependent

Multiple	Bytes	Power of 2
Kilobyte (KB)	1,024	2^{10}
Megabyte (MB)	1,048,576	2^{20}
Gigabyte (GB)	1,073,741,824	2^{30}
Terabyte (TB)	$\approx 1.1 \times 10^{12}$	2^{40}
Petabyte (PB)	—	2^{50}
Exabyte (EB)	—	2^{60}

Trap: disk manufacturers use decimal ($1 \text{ GB} = 10^9 \text{ bytes}$), operating systems use binary (2^{30}). That is why a "500 GB" drive shows as 465 GB. In an exam, **1 KB = 1024 bytes** unless the question says otherwise.

Second trap: KB is a unit of *storage*; **Kbps is a unit of speed** (kilobits per second).

2024 asked "which is the odd one out — KB, MB, Word, KBPS" and the intended answer was

Word (the only one that is not a size unit) — though KBPS is defensible. [△](#) verify.

Coding schemes

Scheme	Bits	Characters	Origin	Notes
BCD (Binary Coded Decimal)	4 bits per decimal digit	0–9 only	—	Each decimal digit is coded separately: 59 → 0101 1001 . 1010–1111 are invalid . Wastes space but makes decimal I/O trivial — used in calculators, digital clocks, financial systems
6-bit BCD	6	64	IBM, early	Uppercase, digits, few symbols
ASCII	7 (extended: 8)	128 (extended: 256)	ANSI, 1963	The one to know cold . The 8th bit was originally a parity bit
EBCDIC	8	256	IBM mainframes	Extended Binary Coded Decimal Interchange Code. Incompatible with ASCII; letters are not contiguous
Unicode	8/16/32 variable	> 149,000	Unicode Consortium, 1991	UTF-8 (1–4 bytes, ASCII-compatible, the web standard), UTF-16 (2 bytes in the BMP), UTF-32 (fixed 4 bytes). Covers every writing system
Gray code	n	2 ⁿ	—	Successive values differ in exactly one bit . Used in shaft encoders and K-maps to avoid transition glitches

ASCII code points you should memorise

Character	Decimal	Hex	Binary
NUL	0	00	0000000
Bell (BEL)	7	07	—
Backspace	8	08	—
Tab (HT)	9	09	—
Line feed (LF)	10	0A	—
Carriage return (CR)	13	0D	—
Escape (ESC)	27	1B	—
Space	32	20	0100000
0	48	30	0110000
9	57	39	—
A	65	41	1000001
Z	90	5A	—
a	97	61	1100001
z	122	7A	—
DEL	127	7F	1111111

Three shortcuts worth knowing:

1. **a - A = 32**. Lowercase is uppercase with bit 5 set. That is why case conversion is one XOR with **0x20**.
2. **The digits start at 48**. So **'7' - '0' = 7** converts a character digit to its value — the standard C idiom.
3. **Codes 0–31 are control characters**, 32–126 are printable, 127 is DEL.

Worked example

Encode the word **Cab** in 7-bit ASCII and in hexadecimal.

C = 67 = 100 0011 = **43**₁₆

a = 97 = 110 0001 = **61**₁₆

b = 98 = 110 0010 = **62**₁₆

So **Cab** = **1000011 1100001 1100010** = **43 61 62** in hex.

Represent 749 in BCD and compare with pure binary.

BCD: 7 → **0111**, 4 → **0100**, 9 → **1001** ⇒ **0111 0100 1001** (12 bits)

Pure binary: 749 = 512+128+64+32+8+4+1 = **1011101101** (10 bits)

BCD needs more bits — that is its cost. Its benefit is that decimal display and input need no conversion arithmetic.

Likely exam questions

- Define bit, byte, nibble and word. [5]
- What is BCD? Represent 428 in BCD. [5]
- Differentiate between ASCII and EBCDIC. [5]
- Explain the various coding schemes used to represent data in a computer. [10]

MCQ traps

- **7-bit ASCII represents 128 characters, and the highest code is 127**. "How many" vs "highest code" differ by one — read the question.
- **1010 is not valid BCD**. Any nibble from 1010 to 1111 is invalid.
- **EBCDIC is 8-bit and IBM's**; ASCII is 7-bit and ANSI's.
- Unicode is not "16-bit" flatly — **UTF-16 is 16 bits in the BMP**. If the option says "Unicode is 16-bit", it is the intended answer on a Diploma paper, but know the nuance.

4. Number systems, conversions and binary arithmetic

Concept

A positional number system is just an agreement about **what each column is worth**. In base 10, the columns from the right are worth 1, 10, 100, 1000 — that is, 10^0 , 10^1 , 10^2 , 10^3 . Change the base and only the column values change; the method is identical. Base 2 columns are worth 1, 2, 4, 8, 16; base 8 columns are 1, 8, 64; base 16 columns are 1, 16, 256.

That is the whole idea. Every conversion below is an application of it.

Why computers use binary: a switch is reliably either on or off, but distinguishing ten distinct voltage levels reliably is hard and expensive. **Why we also use octal and hex:** binary is correct but unreadable — **11011110** is hard for a human to hold in mind, **DE** is easy — and the conversion is trivial because $8 = 2^3$ and $16 = 2^4$, so digits group perfectly.

System	Base	Digits	Group size in binary
Binary	2	0, 1	—

Octal	8	0–7	3 bits
Decimal	10	0–9	—
Hexadecimal	16	0–9, A–F (A=10 ... F=15)	4 bits

The 4-bit table. Memorise it and half of all conversion questions become instant.

Dec	Bin	Oct	Hex		Dec	Bin	Oct	Hex
0	0000	0	0		8	1000	10	8
1	0001	1	1		9	1001	11	9
2	0010	2	2		10	1010	12	A
3	0011	3	3		11	1011	13	B
4	0100	4	4		12	1100	14	C
5	0101	5	5		13	1101	15	D
6	0110	6	6		14	1110	16	E
7	0111	7	7		15	1111	17	F

And the powers of 2: **1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536**. Know them to 2^{16} without thinking.

4.1 Conversions — every direction, worked

(a) Any base → decimal: multiply each digit by its column weight

$(1101101)_2 \rightarrow$ decimal

Bit	1	1	0	1	1	0	1
Weight	64	32	16	8	4	2	1
Product	64	32	0	8	4	0	1

$$64 + 32 + 8 + 4 + 1 = 109_{10}$$

$(357)_8 \rightarrow$ decimal

$$3 \times 8^2 + 5 \times 8^1 + 7 \times 8^0 = 3(64) + 5(8) + 7 = 192 + 40 + 7 = 239_{10}$$

$(2AF)_{16} \rightarrow$ decimal

$$2 \times 256 + 10 \times 16 + 15 \times 1 = 512 + 160 + 15 = 687_{10}$$

(b) Decimal → any base: repeated division, read remainders upwards

156 → binary

Division	Quotient	Remainder
$156 \div 2$	78	0
$78 \div 2$	39	0
$39 \div 2$	19	1
$19 \div 2$	9	1

$9 \div 2$	4	1
$4 \div 2$	2	0
$2 \div 2$	1	0
$1 \div 2$	0	1

Read the remainder column **bottom to top**: $(10011100)_2$

Check: $128 + 16 + 8 + 4 = 156$ ✓

156 → octal

$156 \div 8 = 19 \text{ r } 4$; $19 \div 8 = 2 \text{ r } 3$; $2 \div 8 = 0 \text{ r } 2 \Rightarrow (234)_8$

Check: $2(64) + 3(8) + 4 = 128 + 24 + 4 = 156$ ✓

156 → hexadecimal

$156 \div 16 = 9 \text{ r } 12 = \text{C}$; $9 \div 16 = 0 \text{ r } 9 \Rightarrow (9\text{C})_{16}$

Check: $9(16) + 12 = 144 + 12 = 156$ ✓

(c) Binary ↔ octal: group in 3s from the binary point

$(10011100)_2 \rightarrow \text{octal}$

Pad to a multiple of 3 from the right: $010 \ 011 \ 100 \rightarrow 2, 3, 4 \rightarrow (234)_8$ ✓ (matches above)

$(357)_8 \rightarrow \text{binary}$

$3 = 011$, $5 = 101$, $7 = 111 \Rightarrow (011101111)_2 = 11101111$

(d) Binary ↔ hexadecimal: group in 4s from the binary point

$(10011100)_2 \rightarrow \text{hex}$

$1001 \ 1100 \rightarrow 9, \text{C} \Rightarrow (9\text{C})_{16}$ ✓

$(2\text{AF})_{16} \rightarrow \text{binary}$

$2 = 0010$, $\text{A} = 1010$, $\text{F} = 1111 \Rightarrow (001010101111)_2 = 1010101111$

(e) Octal ↔ hexadecimal: always go via binary

$(357)_8 \rightarrow \text{hex}$

Step 1, to binary in 3s: $011 \ 101 \ 111 = 011101111$

Step 2, regroup in 4s from the **right**, padding left: $0000 \ 1110 \ 1111$

Step 3: 0, E, F $\Rightarrow (\text{EF})_{16}$

Check: $357_8 = 239_{10}$; $\text{EF}_{16} = 14(16) + 15 = 239$ ✓

(f) Fractions: multiply by the base, read integer parts downwards

$0.6875_{10} \rightarrow \text{binary}$

Step	Product	Integer part
0.6875×2	1.375	1
0.375×2	0.75	0
0.75×2	1.5	1
0.5×2	1.0	1
fraction = 0, stop		

Read **top to bottom**: $(0.1011)_2$

Check: $1/2 + 0 + 1/8 + 1/16 = 0.5 + 0.125 + 0.0625 = 0.6875$ ✓

$(1101.101)_2 \rightarrow$ decimal

Integer: $8 + 4 + 1 = 13$. Fraction: $1/2 + 0 + 1/8 = 0.625$. $\Rightarrow 13.625_{10}$

$0.1_{10} \rightarrow$ binary is **non-terminating**: $0.0001100110011\dots$ (recurring 0011). This is why floating-point arithmetic cannot represent 0.1 exactly. Worth one line in a 15-marker.

4.2 Binary arithmetic

Addition — four rules

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1
1+1+1		1	1

1011 + 1101

...

1 1 1 $\leftarrow$ carries

1 0 1 1 (11)

+ 1 1 0 1 (13)

1 1 0 0 0 (24) ✓

...

Subtraction by borrowing — four rules

A	B	Difference	Borrow
0	0	0	0
1	0	1	0
1	1	0	0
0	1	1	1 (borrow 1 from the next column)

$(1011.001)_2 - (110.10)_2$ (this exact sum appeared in the 2024 paper)

Align the binary points and pad: $1011.001 - 0110.100$

...

1 0 1 1 . 0 0 1

- 0 1 1 0 . 1 0 0

0 1 0 0 . 1 0 1

Working right to left: $1-0 = 1$; $0-0 = 0$; $0-1 \rightarrow$ borrow, $= 1$ and the 1 in the units column becomes 0 ; then $0-0 = 0$ (units), $1-1 = 0$ (twos), $0-1 \rightarrow$ borrow from the 8s: $= 1$, and 1 becomes 0 .

Result **$(100.101)_2$**

Check: $1011.001_2 = 11.125$; $110.10_2 = 6.5$; $11.125 - 6.5 = 4.625 = 100.101_2$ ✓

4.3 Complements — and why they exist

The intuition first. Building an adder circuit is easy; building a *separate* subtractor is extra hardware and extra cost. So designers found a trick: **express subtraction as addition of a complement**. $A - B$ becomes $A + (\text{complement of } B)$. One circuit does both operations. Every CPU in existence uses this.

	1's complement	2's complement
How to form	Invert every bit ($0 \leftrightarrow 1$)	1's complement, then add 1
Representations of zero	Two ($+0 = 00000000$, $-0 = 11111111$)	One (00000000)
Range for n bits	$-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$	-2^{n-1} to $+(2^{n-1} - 1)$
8-bit range	-127 to +127	-128 to +127
Carry handling in subtraction	End-around carry — add the carry back in	Discard the carry
Used in modern hardware	Rarely	Universally

Shortcut for forming a 2's complement by inspection

Scan from the **right**: copy bits up to and **including the first 1**, then **invert everything to the left of it**.

2's complement of `00101100`: rightmost bits are `...100`. Copy `100`, invert the rest

(`00101` $\rightarrow$ `11010`) $\Rightarrow$ **`11010100`**.

Check the long way: invert all = `11010011`, add 1 = `11010100` ✓

Worked: 1's complement subtraction

Compute $45 - 27$ using 8-bit 1's complement.

`45 = 00101101`, `27 = 00011011`

1's complement of `27 = 11100100`

...

`00101101` (45)

+ `11100100` (1's comp of 27)

`100010001` $\leftarrow$ carry out = 1

` ` ****There is a carry out, so the result is positive: add the carry back in (end-around carry).** `**00010001 + 1 = **00010010` = 18**` ✓

Compute $27 - 45$ using 8-bit 1's complement.

1's complement of `45 = 11010010`

...

`00011011` (27)

+ `11010010` (1's comp of 45)

`011101101` $\leftarrow$ no carry out

` ` ****No carry out, so the result is negative and is in 1's complement form.**** Take the 1's complement of the result to get the magnitude: `11101101 - 00010010` = 18.`

Answer = **-18** ✓

Worked: 2's complement subtraction

Compute $45 - 27$ using 8-bit 2's complement.

2's complement of 27 = $11100100 + 1 = 11100101$

...

0 0 1 0 1 1 0 1 (45)

+ 1 1 1 0 0 1 0 1 (2's comp of 27)

1 0 0 0 1 0 0 1 0 ← carry out

`` **Carry out = 1 ⇒ result positive; DISCARD the carry.** Answer = 00010010` = 18 ✓

Compute 27 – 45 using 8-bit 2's complement.

2's complement of 45 = $11010010 + 1 = 11010011$

...

0 0 0 1 1 0 1 1 (27)

+ 1 1 0 1 0 0 1 1 (2's comp of 45)

0 1 1 1 0 1 1 1 0 ← no carry out

`` **No carry out ⇒ result is negative and in 2's complement form.** Take the 2's complement of the result: invert 11101110 → 00010001, add 1 → 00010010` = 18.

Answer = -18 ✓

The rule, in one box — memorise this

```
+-----+
| 1's COMPLEMENT SUBTRACTION (A - B) |
| 1. Form the 1's complement of B (invert all bits) |
| 2. Add it to A |
| 3. CARRY OUT? -> result POSITIVE. ADD the carry back in |
|               (END-AROUND CARRY). That is the answer. |
| NO CARRY? -> result NEGATIVE. Take the 1's complement |
|               of the sum; prefix a minus sign. |
+-----+
| 2's COMPLEMENT SUBTRACTION (A - B) |
| 1. Form the 2's complement of B (invert all bits, add 1) |
| 2. Add it to A |
| 3. CARRY OUT? -> result POSITIVE. DISCARD the carry. |
| NO CARRY? -> result NEGATIVE. Take the 2's complement |
|               of the sum; prefix a minus sign. |
+-----+
| MEMORY HOOK: 1's complement -> "1 comes back" (end-around) |
|               2's complement -> "throw it away" |
+-----+
```

Signed number representations compared

Decimal	Sign-magnitude (8-bit)	1's complement	2's complement
+5	0000 0101	0000 0101	0000 0101
-5	1000 0101	1111 1010	1111 1011
+0	0000 0000	0000 0000	0000 0000
-0	1000 0000	1111 1111	<i>does not exist</i>
-128	<i>not representable</i>	<i>not representable</i>	1000 0000

In all three, the **MSB is the sign bit: 0 = positive, 1 = negative.**

Likely exam questions

- Convert $(1101101)_2$ to decimal, octal and hexadecimal. [5]
- Subtract 27 from 45 using (a) 1's complement and (b) 2's complement arithmetic. [5]
- What is 2's complement? Why is it preferred over 1's complement? [5]
- Explain the number systems used in computers and demonstrate all conversions with examples. [15]

MCQ traps

- **1's vs 2's complement.** If the question says "2's complement", the "+1" is not optional. This is the single most common careless error in the whole syllabus.
- **End-around carry belongs to 1's complement only.** In 2's complement you discard.
- Octal ↔ hex **cannot** be done directly by grouping; you must pass through binary.
- Padding: when grouping binary into 3s or 4s, **pad on the left for the integer part and on the right for the fraction part.** Getting this backwards silently changes the answer.

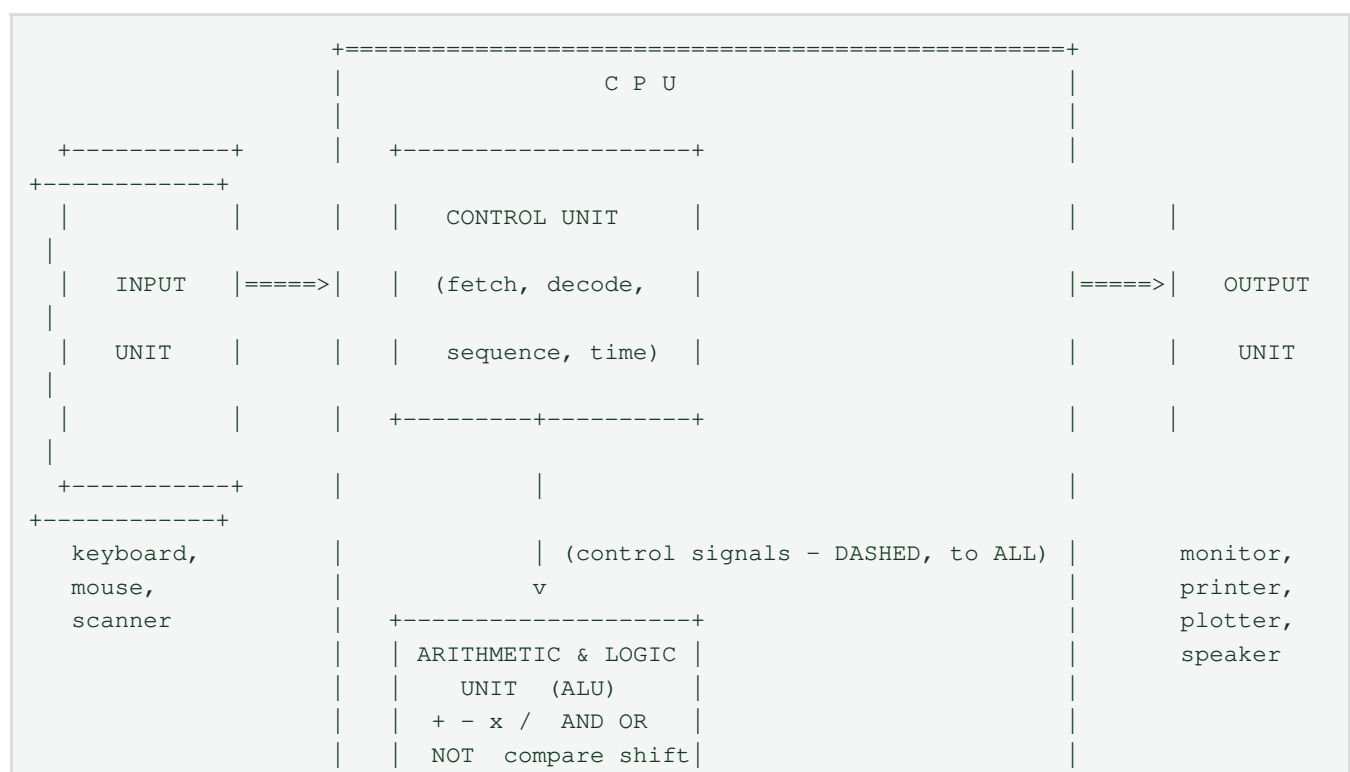
5. Basic organisation of a computer

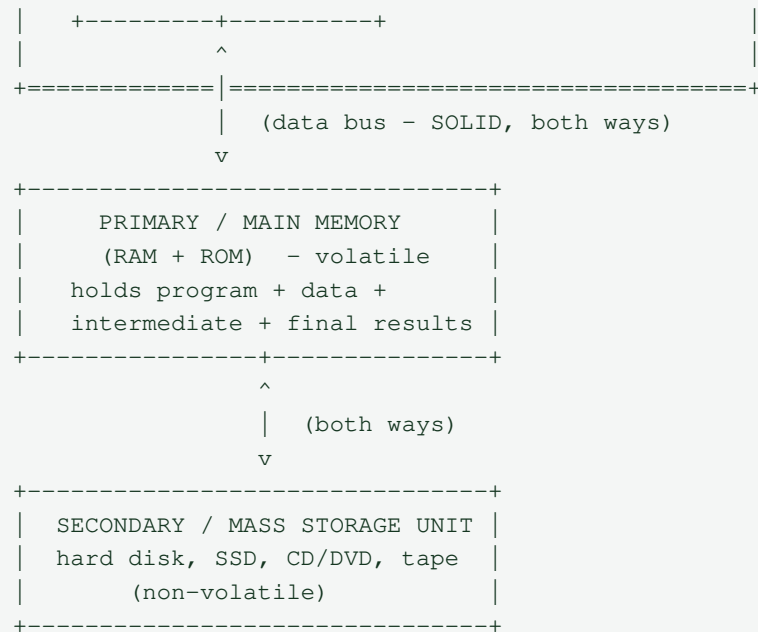
Concept

Strip away every detail and a computer is five boxes. Something has to **get data in**, something has to **hold it**, something has to **do arithmetic on it**, something has to **decide what happens next**, and something has to **get results out**. That is the whole architecture, and it has not changed since von Neumann described it in 1945.

The key insight von Neumann added — the **stored-program concept** — is that the *program* lives in the same memory as the *data*. Before that, "programming" meant rewiring the machine by hand. Once instructions are just numbers in memory, a machine can be re-purposed in seconds, and a program can even modify itself.

The functional block diagram — draw it exactly like this





LEGEND: ==> or <--> solid line = DATA / INSTRUCTION flow
 - - - > dashed line = CONTROL signals from the CU to every block

Drawing instructions for the exam. Draw the CPU as one large box containing two inner boxes (CU on top, ALU below). Put Input on the left, Output on the right, Memory below the CPU, and Secondary Storage below Memory. Use **solid arrows** for the data path and **dashed arrows** radiating from the Control Unit to *every* other block. Label the dashed arrows "control signals" once. Add "CPU = CU + ALU + Registers" as a caption. That caption alone earns a mark.

The five units

Unit	Function	Examples
Input unit	Accepts data and instructions from the outside world; converts them to binary ; supplies them to main memory	Keyboard, mouse, scanner, microphone
Memory unit (primary)	Stores the program, the input data, intermediate results and the final results before output. Directly accessible by the CPU. Volatile	RAM, ROM, cache
ALU	Performs arithmetic (+ – × ÷) and logic (AND, OR, NOT, XOR, compare, shift) operations. Works with the accumulator and the flag/status register	—
Control unit	The nerve centre. Fetches each instruction, decodes it, generates the timing and control signals that make every other unit act, and sequences the next instruction. Processes no data itself	—
Output unit	Converts binary results into human-readable form and presents them	Monitor, printer, speaker, plotter
Secondary / mass storage	Permanent bulk storage of programs and data not currently in use. Non-volatile , cheaper and slower than main memory, not directly addressable by the CPU	Hard disk, SSD, DVD, tape

CPU = Control Unit + ALU + Registers. The CPU is also called the **microprocessor** when it is on one chip, and is described as the **"brain" of the computer**. Within the CPU, the **ALU is sometimes called the "heart" of the processor** (2024 asked exactly this).

The machine cycle (fetch–decode–execute–store)

Step	Who does it	What happens
1. Fetch	CU	The instruction at the address in the PC is copied from memory into the IR
2. Decode	CU	The instruction is interpreted; operands are identified
3. Execute	ALU	The operation is performed
4. Store	CU	The result is written back to a register or to memory

The PC is then incremented (or loaded with a branch target) and the cycle repeats.

Steps 1–2 are the *instruction cycle*; 3–4 are the *execution cycle*.

2024 trap: "Which is **not** a part of the CPU cycle — fetching, decoding, controlling, storing?" The intended answer is **controlling** (it is what the CU *is*, not a phase).

Important registers

Register	Full name	Holds
PC	Program Counter	Address of the next instruction to be executed
IR	Instruction Register	The instruction currently being executed
MAR	Memory Address Register	The address of the memory location to be read from or written to
MBR / MDR	Memory Buffer / Data Register	The data being transferred to or from memory
AC	Accumulator	The operand and result of ALU operations
Flags / PSW	Status register	Zero, carry, sign, overflow, parity bits
SP	Stack Pointer	Address of the top of the stack

PC vs MAR is a favourite MCQ. PC = *next instruction's* address. MAR = address of *whatever memory location is about to be accessed*, instruction or data.

MBR is not a register type — beware: in a motherboard context MBR means **Master Boot Record**. 2024 offered "PC, PCB, MAR, MBR" for "which is not a type of register" and the intended answer was **PCB** (Process Control Block, an OS data structure).

Likely exam questions

- What are the basic functional units of a computer? Explain the role of each. **[5]**
- Explain, with a labelled block diagram, the basic organisation of a computer. **[10]**
- Explain the machine cycle. Name the registers involved and state what each holds. **[10]**

6. Input devices

6.1 Keyboard

Concept. A keyboard is a grid of switches. Pressing a key closes a connection at one row–column intersection; a small microcontroller inside the keyboard scans the grid continuously, detects which intersection closed, and sends a **scan code** — not an ASCII code — to the PC.

The keyboard controller and the OS then translate the scan code into a character, applying Shift, Caps Lock, and the current keyboard layout. There is a separate scan code for *key-down* and *key-up*, which is how the machine knows you are holding Shift.

Feature	Detail
Standard layout	QWERTY (named after the first six top-row letters). Alternatives: DVORAK, AZERTY (French), COLEMAK
Key count	101 keys (enhanced/AT-101); 104 keys with the two Windows keys and the menu key. Older PC/XT keyboard = 83/84 keys
Function keys	F1 – F12 → twelve (asked in 2024)
Key groups	Alphanumeric (typewriter) · Numeric keypad (17 keys) · Function keys (F1–F12) · Cursor/navigation (arrows, Home, End, PgUp, PgDn, Ins, Del) · Modifier/special (Shift, Ctrl, Alt, Caps Lock, Num Lock, Scroll Lock, Esc, Tab, Enter, Backspace, PrtScn , Pause)
Construction types	Membrane (cheap, quiet, mushy) · Mechanical (individual switches, tactile, durable, ~50 M keystrokes) · Capacitive · Scissor-switch (laptops)
Other types	Ergonomic/split, wireless (RF or Bluetooth), virtual/on-screen, projection, gaming, multimedia, Braille
Interfaces	PS/2 (purple , 6-pin mini-DIN) → USB → Bluetooth

A keyboard is not a pointing device. It can *substitute* for a mouse (Tab, arrow keys, keyboard shortcuts, Sticky Keys) but it does not control 2-D cursor movement directly. 2024 tested exactly this distinction.

6.2 Mouse

Concept. A mouse reports **relative displacement**, not absolute position. It says "I moved 40 units right and 12 up since I last reported"; the OS adds that to the pointer's current position. That is why lifting a mouse and putting it down elsewhere does not teleport the pointer.

Type	How it senses motion	Pros	Cons
Mechanical / ball	A rubber-coated steel ball rolls against two perpendicular rollers (X and Y); each roller turns a slotted encoder wheel interrupting an LED–photodetector pair; pulse counts give displacement	Cheap, works on any surface	Ball picks up dirt; rollers clog; needs cleaning; lower precision; moving parts wear
Optical	An LED (typically red) illuminates the surface at a shallow angle; a small CMOS image sensor photographs it 1,500–6,000+ times per second; an on-board DSP cross-correlates successive images to compute Δx and Δy	No moving parts, no cleaning, higher precision, more reliable	Struggles on glass and uniform glossy surfaces
Laser	Same as optical but uses a laser diode (VCSEL) instead of an LED — a coherent source resolves finer surface detail	Highest precision (up to 20,000 DPI); works on more surfaces including glossy	Costlier; can be <i>too</i> sensitive
Trackball	An inverted mouse — the user rolls the ball directly; the device stays still	Needs no desk space; good for RSI	Different motor skill

Touchpad	Capacitive sensing grid detects finger position	Integrated in laptops	Less precise for fine work
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Parts of an optical mouse: LED (or laser diode), **lens**, **optical/CMOS image sensor**, DSP chip, left/right buttons, **scroll wheel**, cable or wireless transceiver, PCB.

2024 trap: "Which is **not** a part of an optical mouse — LED, optical sensor, CCD, diode?" The intended answer is **CCD** — optical mice use **CMOS** sensors, not CCDs.

The scroll wheel is a third axis: a notched wheel between the buttons, read by its own encoder, generating scroll events; pressing it acts as a middle-button click.

Mouse resolution is quoted in **DPI (dots per inch)** or CPI (counts per inch): higher DPI means the pointer travels further on screen for the same physical movement.

Mouse operations to name in an answer: point, click, double-click, right-click (context menu), **drag and drop**, scroll, hover.

6.3 Scanners

Concept. A scanner is a camera that moves. A bright lamp illuminates the document; the light reflected from it is directed by mirrors and a lens onto a row of light-sensitive cells (a **CCD array**); the assembly traverses the page line by line; each cell's output voltage — proportional to the brightness it sees — is digitised by an **ADC** into a pixel value. Colour is captured by using three filters (red, green, blue) or three sensor rows.

Type	Description	Use
Flatbed	Document laid on a glass platen; the scan head moves beneath	General office; the most common
Sheet-fed	The document moves past a fixed head	Multi-page documents, fax machines
Handheld	Dragged over the document by hand	Small images, portable use
Drum	Document mounted on a rotating drum, read by a photomultiplier tube	Professional pre-press; the highest quality
3-D scanner	Captures object geometry by laser or structured light	CAD, forensics, printing

Scan parameters you set before scanning: **resolution (DPI)**, **colour mode** (line art / greyscale / colour) and **colour depth** (bits per pixel), **document size**, **file format**, brightness/contrast, and **simplex vs duplex** for sheet-fed units.

Note: "scanning method" is not a normally-set parameter — 2024 used it as the odd one out.

6.4 OCR, OMR, MICR and bar-code readers — the four "reader" technologies

These four are constantly confused and therefore constantly examined. Learn the table.

	OCR	OMR	MICR	Bar-code reader
Full name	Optical Character Recognition	Optical Mark Recognition/Reader	Magnetic Ink Character Recognition	—

What it reads	Printed or typed characters , converting the image to editable text	The presence or absence of a mark in a predefined position	Characters printed in magnetic (iron-oxide) ink	A pattern of parallel bars and spaces of varying width
How it works	Scan the page → segment into characters → match each against stored patterns / neural network → output character codes	Reflected-light sensors detect darkened bubbles at fixed coordinates	The document passes a magnetising head , then a read head detects the magnetic signature of each character shape	A light source scans across; dark bars absorb, light spaces reflect ; a photodiode converts the reflection to a pulse train; a decoder converts it to digits
Typeface	Any (with training)	n/a	E-13B (US/India) or CMC-7 (Europe)	UPC, EAN-13, Code 39, Code 128, QR (2-D)
Accuracy	Good but imperfect; errors on poor scans and handwriting	Very high	Extremely high — unaffected by stamps, folds, signatures over the characters	Very high
Typical use	Digitising books, forms, number plates, passports	OMR answer sheets for competitive exams (this exam's own answer sheet!) , ballot papers, surveys	Bank cheques — the code line at the bottom	Retail point of sale, inventory, library issue, courier tracking, boarding passes

Bar-code structure worth naming: quiet zone, start character, data characters, **check digit**, stop character, quiet zone. The **check digit** validates the read.

6.5 Other input devices — one line each

Device	Description
Joystick	A pivoted stick reporting direction and displacement; gaming, flight simulators, industrial control
Light pen	A photodetector in a stylus; detects when the CRT's electron beam passes the point it is touching, giving its position
Touch screen	A display overlaid with position sensing — resistive (two conductive layers pressed together, works with any stylus), capacitive (senses the finger's charge, supports multi-touch), infrared, surface-acoustic-wave. Both an input and an output device
Digitiser / graphics tablet	A flat pad plus a stylus; reports absolute position; used for drawing, CAD, signature capture
Microphone	Converts sound pressure into an analog electrical signal, digitised by the sound card's ADC
Webcam / digital camera	Image sensor (CMOS/CCD) plus lens; captures still or moving images
MIDI keyboard	Sends note/velocity messages, not audio
Biometric devices	Fingerprint, iris, retina, face, voice, palm-vein readers
Smart card / magnetic stripe reader	Reads a chip or magnetic stripe

RFID / NFC reader	Reads a tag by radio; NFC allows a phone to act as a contactless smart card
Sensors	Temperature, pressure, light, accelerometer, GPS

Likely exam questions

- Explain the working of an optical mouse. How does it differ from a mechanical mouse? [5]
- Differentiate between OCR, OMR and MICR. [5]
- What is a bar-code reader? Explain its working and applications. [5]
- Explain the various input devices of a computer. [10]

MCQ traps

- **MICR = cheques. OMR = answer sheets. OCR = converting scanned text.** Do not mix them.
- **Touch screen, modem, NIC and a multifunction printer are BOTH input and output devices.**
- An optical mouse uses a **CMOS** sensor, not a CCD.
- **A scanner is an input device; a plotter is an output device.** Obvious until you are tired.

7. Output devices

7.1 Monitors — CRT vs LCD

How a CRT works

Inside an evacuated glass tube, an **electron gun** heats a cathode until it emits electrons. A high positive voltage on the anode accelerates them into a beam. **Deflection coils** (magnetic) around the neck bend the beam so it sweeps across the screen in a raster — left to right, top to bottom. The inside of the screen face is coated with **phosphor**, which glows where the beam strikes it. Beam intensity controls brightness. For colour, there are **three guns (R, G, B)** and three phosphor dots per pixel, with a **shadow mask** or **aperture grille** ensuring each gun's beam hits only its own colour of phosphor.

Because the phosphor glow decays, the whole screen must be **redrawn many times per second** — the **refresh rate** (60–100 Hz). Too low a refresh rate produces visible flicker and eye strain.

How an LCD works

Liquid crystals are molecules that **twist the polarisation of light**, and the amount of twist changes with an applied voltage. An LCD panel is a sandwich: **backlight** (CCFL or LED) → **polariser** → glass with transparent electrodes → **liquid crystal layer** → colour filter → second polariser at 90°. With no voltage, the crystals twist the light 90° so it passes the second polariser — the pixel is bright. Apply voltage and the crystals untwist, so the light is blocked — the pixel is dark. Intermediate voltages give intermediate brightness. Each pixel has three sub-pixels with red, green and blue filters.

TFT (Thin Film Transistor) / active matrix gives each sub-pixel its own transistor and capacitor, so it holds its state; **passive matrix** addresses rows and columns in sequence and is slower and lower-contrast.

An LCD emits no light of its own — it modulates a backlight. That is why an LCD washes out in sunlight and why "LED monitor" is a marketing term for *an LCD with an LED backlight*.

The comparison table

Feature	CRT	LCD
Technology	Electron beam on phosphor	Liquid crystals modulating a backlight
Physical depth / weight	Very deep, very heavy	Thin, light
Power consumption	High (70–150 W)	Low (20–40 W)
Radiation / emissions	Emits X-rays and EM radiation	Negligible
Flicker	Present at low refresh rates	None — pixels hold state
Geometric distortion	Possible (pincushion, barrel)	None — a fixed pixel grid
Resolution	Any resolution looks acceptable (multisync)	Sharp only at the native resolution ; scaled images blur
Colour reproduction	Excellent; wide gamut, deep blacks	Very good; older panels had poorer blacks and gamma
Contrast ratio	Very high (true black when the beam is off)	Lower on basic panels (backlight leaks)
Viewing angle	Wide	Narrower on TN panels; wide on IPS
Response time	Instant	Slower (ghosting on early panels)
Screen size vs footprint	Small screen, huge footprint	Large screen, tiny footprint
Heat	High	Low
Cost	Cheaper historically	Cheaper now; CRT is obsolete
Lifetime	Phosphor burn-in possible	Backlight dims over time

Monitor terminology to define in an answer: resolution (e.g. 1920 × 1080), **pixel**, **dot pitch** (distance between same-colour phosphor dots — smaller is sharper), refresh rate (Hz), aspect ratio (4:3, 16:9), colour depth (bits/pixel), **VDU** (Visual Display Unit, the formal name), video adapter/graphics card, and the connectors VGA (analog), DVI, HDMI, DisplayPort.

Also mention for completeness: **LED** (LCD with LED backlight, or true OLED), **plasma** (each cell is a tiny fluorescent lamp of ionised gas — big, bright, power-hungry, obsolete), and **OLED** (organic LEDs emit their own light — perfect blacks, thin, flexible).

7.2 Printers

The primary division

	Impact printers	Non-impact printers
Mechanism	A print head physically strikes an inked ribbon against the paper	No physical contact ; ink is sprayed, or toner is fused, or paper is heated
Noise	Loud	Quiet
Multi-part / carbon copies	YES — the only kind that can	No

Speed	Slow	Fast
Quality	Lower	Higher
Examples	Dot matrix , daisy wheel, line printer, drum printer, chain printer	Laser , inkjet, thermal, electrostatic

How each of the three works

Dot matrix (DMP). The print head carries a vertical column of **9 or 24 tiny pins** driven by solenoids. As the head traverses the line, the right combination of pins fires at each horizontal position, striking an inked ribbon and pressing ink onto the paper. Characters are therefore built from a **matrix of dots** — hence the name. 24-pin heads give finer dots and "near letter quality" (NLQ). Uses continuous **tractor-feed** stationery.

Inkjet. A print head with dozens to thousands of microscopic nozzles sprays **droplets of liquid ink** (picolitres) directly at the paper. Two ejection technologies: **thermal (bubble-jet)** — a tiny heater vaporises a bubble of ink which ejects a droplet; and **piezoelectric** — a crystal deforms under voltage and squeezes out a droplet. Colour comes from CMYK cartridges combining on the page.

Laser. A **page printer**, and the mechanism is worth knowing as a sequence of seven steps:

1. **Charging** — a corona wire or charge roller gives the photosensitive **drum** a uniform negative charge.
2. **Writing/exposing** — a **laser beam**, steered by a rotating polygonal mirror, scans the drum, discharging the spots it hits. This writes a hidden **electrostatic latent image**.
3. **Developing** — the drum rolls past the **toner** (a fine plastic-and-pigment powder), which sticks only to the discharged areas.
4. **Transferring** — the paper, given an opposite charge, passes the drum and pulls the toner off.
5. **Fusing** — hot rollers (~200 °C) **melt the toner into the paper fibres**. This is why laser output comes out warm and is smudge-proof.
6. **Cleaning** — a blade scrapes residual toner off the drum.
7. **Discharging** — an erase lamp neutralises the drum for the next page.

The three-way comparison table — learn this cold

Feature	Dot Matrix	Inkjet	Laser
Impact / non-impact	Impact	Non-impact	Non-impact
Print element	9 or 24 pins striking a ribbon	Nozzles spraying ink droplets	Laser + drum + toner + heat fusing
Prints by	Character	Character/band	Whole page at once
Consumable	Inked ribbon	Ink cartridge	Toner cartridge + drum
Speed measure	CPS (characters/sec)	PPM	PPM
Typical speed	30 – 600 CPS	4 – 20 PPM	8 – 100+ PPM
Resolution	Low — 72–240 DPI, visible dots	600 – 4800 DPI	600 – 2400 DPI
Text quality	Poor to fair (NLQ on 24-pin)	Good	Excellent — sharpest text

Photo/colour quality	Very poor	Excellent	Good
Noise	Very noisy	Quiet	Quiet
Initial cost	Low	Lowest	High
Cost per page	Lowest	Highest (cartridges)	Low to medium
Carbon / multi-part copies	YES	No	No
Ink smudging	No	Possible until dry	No (fused)
Duty cycle	Rugged; tolerates dust and heat	Light	Heavy
Best suited to	Invoices, bills, railway tickets, multi-part stationery, harsh industrial environments	Home use, photographs, low-volume colour	Offices, high-volume text, sharp documents

Other printer types worth one line

Type	Note
Daisy wheel	Impact; a wheel of fully-formed characters struck by a hammer — typewriter quality but text only, no graphics ; obsolete
Line printer (drum, chain, band)	Impact; prints a whole line at once at 300–3000 LPM ; used with mainframes
Thermal	Non-impact; heats special heat-sensitive paper — receipts, ATM slips, fax. Cheap, but the print fades
Thermal wax / dye-sublimation	Non-impact; heats a coloured wax or dye ribbon — photo-quality continuous tone
3-D printer	Additive manufacturing; builds objects layer by layer

Printer classification by output unit: character printers (DMP, daisy wheel, inkjet) · **line printers** (drum, chain, band) · **page printers** (laser). This three-way split is itself an examinable classification.

7.3 Plotters

Concept. A printer thinks in **dots**: it fills a raster grid. A plotter thinks in **lines**:

it is given the coordinates of the endpoints and physically moves a pen from one to the other, drawing a continuous stroke. That makes a plotter ideal for line drawings at very large sizes — an architectural plan, a circuit layout, a map — where a raster image would need an impossible number of dots, and where line quality matters more than filled colour.

Type	Construction
Drum plotter	The paper is wrapped round a rotating drum that moves it back and forth on one axis; the pen carriage moves left–right on the other axis. Compact for a given paper width; can handle long continuous rolls
Flatbed plotter	The paper lies flat and stationary ; the pen moves in both X and Y on a gantry. Higher accuracy; the machine must be as large as the paper

Inkjet / electrostatic plotter	Modern "plotters" are really large-format inkjet printers; faster and can fill colour
Cutting plotter	The pen is replaced by a blade — vinyl signage, stickers, fabric patterns

	Printer	Plotter
Image built from	Dots (raster)	Continuous lines (vector)
Input	Bitmap / page description	Coordinate pairs / vector commands
Strength	Text and photographs	Line drawings, at very large size
Typical size	A4 / A3	A1, A0 and larger; rolls
Speed	Fast	Slow
Applications	Documents, photos	CAD, engineering drawings, architectural plans, maps, circuit diagrams, banners

Projector (also examinable as an output device): its components are the **optical system** (lens, mirrors), the **display/imaging element** (LCD panel, DLP micromirror chip, or three CRT tubes in older units) and the **light source** (lamp or laser).

Likely exam questions

- *Differentiate between CRT and LCD monitors. [5]*
- *Explain the working of a laser printer. [5] or [10]*
- *Compare dot matrix, inkjet and laser printers. [10]*
- *What is a plotter? How does it differ from a printer? Describe its types. [5]*
- *Explain the various output devices of a computer. [10]*

MCQ traps

- **"Which printer produces carbon copies?"** → **dot matrix**, always, because it is the impact one.
- **DMP speed is in CPS; laser and inkjet in PPM; line printers in LPM.** DPI is *resolution*, not speed.
- **Laser is a page printer;** inkjet and DMP are character printers.
- **A plotter draws lines, a printer prints dots.** If the question mentions engineering drawings or A0 paper, the answer is plotter.
- **An LCD does not emit light;** it modulates a backlight.

8. Storage devices

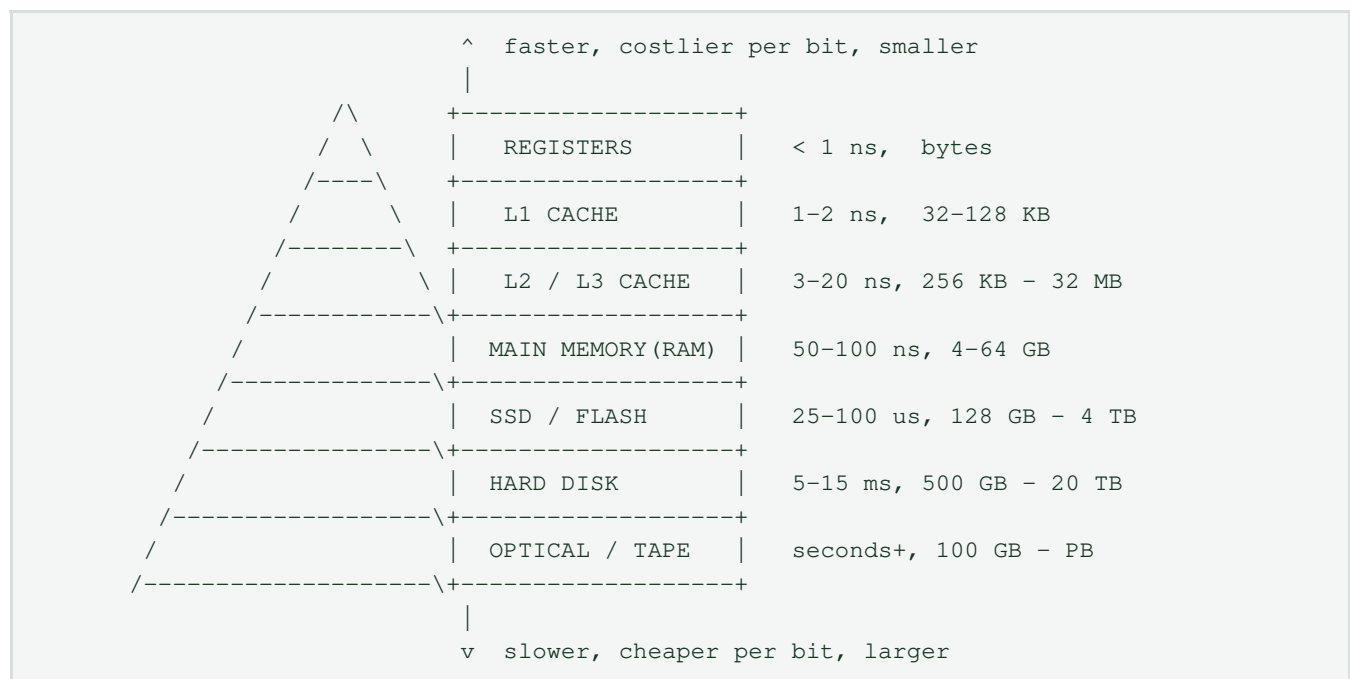
8.1 The organising idea

You cannot have storage that is simultaneously **fast, large and cheap** — pick two. Fast memory (SRAM) costs enormously per bit, so you can only afford a little. Cheap storage (disk) is huge but glacially slow. The engineering answer is not to choose, but to **arrange several technologies in a hierarchy** and let the fast small ones hold whatever is being used right now.

This works because of the **principle of locality**: programs do not access memory randomly. They re-use the same variables (**temporal locality**) and walk through neighbouring addresses (**spatial locality**). So a small fast cache holding the recently-used data captures the great majority of accesses.

Deeper treatment of cache organisation, mapping, replacement and virtual memory is in [CORE_02_ComputerOrganization.md](#). What follows is the Diploma-level treatment: the device facts, the comparison tables, and the numbers.

8.2 The memory hierarchy



Level	Technology	Typical capacity	Access time	Cost/bit	Managed by	Volatile?
Registers	SRAM cells in the CPU	Tens–hundreds of bytes	< 1 ns	Highest	Compiler / CPU	Yes
L1 cache	SRAM	32 – 128 KB	1 – 2 ns	Very high	Hardware	Yes
L2 / L3 cache	SRAM	256 KB – 32 MB	3 – 20 ns	High	Hardware	Yes
Main memory	DRAM	4 – 64 GB	50 – 100 ns	Medium	Operating system	Yes
SSD	NAND flash	128 GB – 4 TB	25 – 100 µs	Low	OS / file system	No
Hard disk	Magnetic	500 GB – 20 TB	5 – 15 ms	Very low	OS / file system	No
Optical / tape	Optical / magnetic	700 MB – PB	Seconds – minutes	Lowest	Operator	No

Get the orders of magnitude right, because a favourite question is "how much slower is disk

than RAM?" Roughly: RAM $\approx$ 100 ns, disk $\approx$ 10 ms. That is a factor of **100,000**. If a register access took one second, a disk access would take about four months.

Effective access time with a cache, worked:

h = hit ratio = 0.9, t_{cache} = 2 ns, t_{main} = 100 ns.

$EAT = h \times t_{\text{cache}} + (1 - h) \times (t_{\text{cache}} + t_{\text{main}}) = 0.9(2) + 0.1(102) = 1.8 + 10.2 = \mathbf{12 \text{ ns}}$.

(Some textbooks use $EAT = h \cdot t_c + (1-h) \cdot t_m = 0.9(2) + 0.1(100) = \mathbf{11.8 \text{ ns}}$. **State which model you are using** — either is accepted if you say so.)

8.3 Primary memory — RAM

	SRAM (Static RAM)	DRAM (Dynamic RAM)
Storage cell	A flip-flop , typically 6 transistors	1 transistor + 1 capacitor
Refresh needed?	No — holds its state while powered	Yes — the capacitor leaks; refreshed every few milliseconds
Speed	Fast (1–10 ns)	Slower (50–100 ns)
Density (bits per unit area)	Low	High
Cost per bit	High	Low
Power consumption	Higher static, no refresh power	Lower static, but refresh consumes power
Complexity of controller	Simple	Needs a refresh circuit
Typical use	Cache memory , registers, small buffers	Main memory
Both are	Volatile	Volatile

The DRAM family — SDRAM vs DDR, and the DDR generations

The step that matters. Early **asynchronous DRAM** responded whenever it could. **SDRAM (Synchronous DRAM)** synchronised the memory to the system clock, so transfers could be pipelined and the CPU knew exactly when data would arrive. **DDR (Double Data Rate) SDRAM** then made one further change: instead of transferring data only on the **rising** edge of the clock, it transfers on **both the rising and the falling** edges — **two transfers per clock cycle**, so roughly double the bandwidth at the same clock frequency.

⚠ **The classic misconception:** DDR does **not** double the clock speed. It doubles the number of transfers per clock. That is exactly what "double data rate" means, and it is the answer the examiner wants.

	SDR SDRAM	DDR SDRAM
Full name	Single Data Rate Synchronous DRAM	Double Data Rate Synchronous DRAM
Transfers per clock cycle	1 (rising edge only)	2 (rising and falling edges)
Bandwidth at the same clock	1×	2×

Prefetch buffer	1 bit	2 bits (DDR1)
Operating voltage	3.3 V	2.5 V
Module pins (DIMM)	168	184
Notches in the module	2	1
Typical speeds	PC66, PC100, PC133	DDR-200 to DDR-400

Generation	Prefetch	Voltage	DIMM pins	Typical data rate
SDR SDRAM	1n	3.3 V	168	66–133 MT/s
DDR	2n	2.5 V	184	200–400 MT/s
DDR2	4n	1.8 V	240	400–1066 MT/s
DDR3	8n	1.5 V	240 (different notch position from DDR2)	800–2133 MT/s
DDR4	8n	1.2 V	288	1600–3200 MT/s
DDR5	16n	1.1 V	288 (different keying)	4800+ MT/s

Each generation **lowers the voltage** and **increases the prefetch depth**. The notch position differs in every generation so a module physically cannot be fitted into the wrong slot — a nice one-line detail to include in an answer.

Naming convention worth knowing: **DDR3–1600** names the **transfer rate in MT/s**;

PC3–12800 names the **peak bandwidth in MB/s** ($1600 \text{ MT/s} \times 8 \text{ bytes} = 12800 \text{ MB/s}$). Same module, two names.

Other RAM terms

Term	Meaning
SIMM	Single Inline Memory Module — contacts on both sides are electrically identical ; 30-pin or 72-pin; obsolete
DIMM	Dual Inline Memory Module — the two sides are independent contacts; 168/184/240/288-pin
SO-DIMM	Small Outline DIMM — the shorter laptop module (200/204/260 pins)
RDRAM	Rambus DRAM — a fast, expensive 1990s alternative that lost to DDR
VRAM / GDDR	Video RAM — dual-ported / high-bandwidth memory on graphics cards
ECC memory	Error-Correcting Code memory — an extra chip stores parity/Hamming bits, detecting and correcting single-bit errors. Used in servers

8.4 Primary memory — the ROM family

Concept. RAM forgets everything when the power goes. But the machine needs *some* code already present at power-on to know how to start — you cannot load the boot program from disk using a program that is itself on disk. So there must be non-volatile memory holding firmware.

That is ROM. The whole family history is one question: **how easily can you change what is in it?**

Type	Full name	Programmed	Erasable?	Erase method	Granularity	Reprogram cycles	Typical use
Mask ROM	Read Only Memory	At manufacture , by a photographic mask	No	—	—	0	High-volume fixed firmware; cheapest at scale
PROM	Programmable ROM	Once, by the user , in a "PROM burner" that blows fuses/ antifuses	No	—	—	1 (OTP)	Small production runs
EPROM	Erasable PROM	Electrically, in a programmer	Yes	Ultraviolet light through a quartz window , ~20 minutes	Whole chip	~100–1000	Development firmware; older BIOS
EEPROM	Electrically Erasable PROM	Electrically	Yes	Electrically, in circuit	Byte by byte	$10^4 - 10^6$	CMOS settings, config data, smart cards
Flash	Flash EEPROM	Electrically	Yes	Electrically	Block / sector	$10^4 - 10^5$	BIOS, SSDs, pen drives, memory cards

All are non-volatile. The progression is a straight line: harder to change → easier to change, and correspondingly the BIOS went from unchangeable to field-updatable software.

NOR vs NAND flash (one line each, worth a mark): **NOR** flash is byte-addressable and supports execute-in-place — used for BIOS. **NAND** flash is page-based, denser and cheaper — used for SSDs, pen drives and memory cards.

8.5 Cache memory

Aspect	Detail
What it is	Small, very fast SRAM between the CPU and main memory
Why	The CPU is ~100× faster than DRAM; without cache it would stall constantly
Why it works	Locality of reference — temporal (same data re-used soon) and spatial (nearby addresses used soon)
Levels	L1 — smallest and fastest, on the CPU core, usually split into instruction (I-cache) and data (D-cache). L2 — larger, per core. L3 — largest, shared between cores

Hit / miss	A hit = the data is in the cache. Hit ratio $h = \text{hits} \div \text{total accesses}$. A well-designed cache achieves $h > 0.9$
Mapping	Direct-mapped · Fully associative · Set-associative (the practical compromise) — see CORE_02
Write policies	Write-through (write to cache and memory together — simple, safe, slower) vs write-back (write only to cache, flush later using a <i>dirty bit</i> — faster, more complex)
Replacement	LRU, FIFO, Random

8.6 Magnetic storage — floppy disks

A **floppy disk** is a thin flexible mylar disk coated with magnetic oxide, spinning inside a protective jacket at ~300 RPM, written and read by heads that **physically touch** the surface (hence the wear and the noise). Data is organised in concentric **tracks** divided into **sectors**, and both surfaces are used.

Memorise all four capacities — this is one of the most reliably asked facts in the syllabus:

Size	Density	Capacity	Tracks/side	Sectors/track	Sides
5.25"	Double density (DD)	360 KB	40	9	2
5.25"	High density (HD)	1.2 MB	80	15	2
3.5"	Double density (DD)	720 KB	80	9	2
3.5"	High density (HD)	1.44 MB	80	18	2
3.5"	Extra-high density (ED)	2.88 MB	80	36	2

Verify the 1.44 MB yourself, because the arithmetic is a common 5-mark question:

$2 \text{ sides} \times 80 \text{ tracks} \times 18 \text{ sectors} \times 512 \text{ bytes} = \mathbf{1,474,560 \text{ bytes}} = 1440 \text{ KB} = \mathbf{1.44 \text{ "MB"}}$

(using the marketing convention $1 \text{ MB} = 1000 \text{ KB}$ — which is why the number is 1.44 and not 1.41).

Physical features of a 3.5" floppy: rigid plastic shell, sliding metal **shutter** protecting the media window, central metal **hub** for the drive spindle, and a **write-protect notch** with a sliding tab — **hole open = write protected**. (On a 5.25" disk it is the opposite: the notch must be *open* to write, so covering it with tape write-protects it. ⚠ Note the reversal.)

Floppy drive components: read/write heads (one per side), head-positioning stepper motor, drive spindle motor, disk-ejection mechanism, index sensor, write-protect sensor, controller electronics. Connected by a **34-pin ribbon cable**.

8.7 Magnetic storage — the hard disk

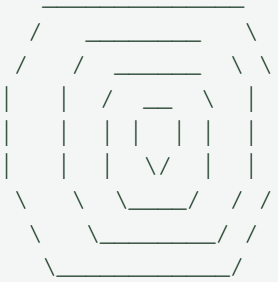
Components and their functions

Component	Function
Platters	Rigid aluminium alloy or glass discs coated on both surfaces with a thin magnetic film. 1–8 per drive. Rigidity is what allows the very close head flying height and hence the high density
Spindle and spindle motor	Holds and rotates all platters as one unit at a constant 5400 / 7200 / 10 000 / 15 000 RPM

Read/write heads	One per platter surface. A tiny electromagnet that magnetises a region to write and senses the field to read. Modern drives use a separate GMR (giant magnetoresistive) element to read
Air bearing / flying height	The heads do not touch the platter — they fly on a cushion of air dragged along by the spinning disk, a few nanometres above it. If a head does touch the surface, that is a head crash and it destroys data
Actuator arm assembly	Carries all the heads; they move together as one unit , which is why the cylinder concept exists
Voice-coil actuator	A magnet-and-coil motor that swings the arm to position the heads. Replaced the older, slower stepper motor
Logic / controller board	The PCB on the underside: interface electronics, cache buffer, servo control, encoding/decoding
HDA (Head-Disk Assembly)	The sealed enclosure containing platters, heads and actuator, kept free of dust
Breather filter	A filtered vent equalising internal and external air pressure. A hard disk is not a vacuum — it needs air for the head to fly on
Landing zone / park position	An unused inner region where the heads rest when powered off

Organisation of data — the four terms

PLATTER (top view)



Concentric circles = TRACKS
Pie-slice divisions = SECTORS
Track x Sector cell = one SECTOR (512 B)

STACK (side view) -- the CYLINDER idea

```

_____ <- head 0
|_____| <- platter 1, head 1
_____ <- head 2
|_____| <- platter 2, head 3
_____ <- head 4
|_____| <- platter 3, head 5

```

The set of tracks at the SAME radius on ALL surfaces = one CYLINDER (a hollow cylinder cut through the stack).

Because all heads move together, reading a whole cylinder needs NO head movement -- which is why the OS allocates by cylinder.

Term	Definition
Track	One concentric circle on one platter surface. Numbered from 0 at the outside edge inwards
Sector	A pie-slice division of a track — the smallest physically addressable unit . Traditionally 512 bytes ; modern "Advanced Format" drives use 4096 bytes . Each sector has a header (address mark), the data field, and an ECC field
Cylinder	The set of all tracks of the same number on all surfaces . Reading a cylinder requires no head movement, so it is the natural allocation unit

Cluster / allocation unit	The smallest unit the file system will allocate to a file — 1, 2, 4, 8 ... sectors. A 1-byte file still consumes one whole cluster; the wasted remainder is slack space (internal fragmentation)
Head	One read/write head per surface; head number identifies the surface
CHS addressing	The legacy Cylinder-Head-Sector scheme; replaced by LBA (Logical Block Addressing) , which numbers every sector linearly from 0
Zone Bit Recording (ZBR)	Outer tracks are physically longer, so they are given more sectors than inner ones. This is why the constant "sectors per track" figure is a simplification, and why sequential transfer is faster at the outside of a modern disk

Capacity and access time — worked

Capacity formula:

$\text{Capacity} = \text{heads} \times \text{cylinders} \times \text{sectors per track} \times \text{bytes per sector}$

Worked example 1. A disk has 4 platters (so 8 surfaces/heads), 16383 cylinders, 63 sectors per track and 512 bytes per sector. Find the capacity.

$8 \times 16383 \times 63 \times 512 = \mathbf{4,227,858,432 \text{ bytes} \approx 4.23 \text{ GB}}$ ($\approx 3.94 \text{ GiB}$).

Worked example 2. A disk has 2 platters, 1024 cylinders, 128 sectors/track, 512 B/sector.

Surfaces = $2 \times 2 = 4$.

Capacity = $4 \times 1024 \times 128 \times 512 = \mathbf{268,435,456 \text{ bytes} = 256 \text{ MiB} = 0.25 \text{ GiB}}$.

Access time formula:

$\text{Access time} = \text{Seek time} + \text{Rotational latency} + \text{Transfer time}$

Component	What it is	Typical
Seek time	Time for the actuator to move the heads to the right cylinder	3 – 15 ms (average ≈ 9 ms)
Rotational latency	Time for the required sector to rotate under the head. On average, half a revolution	See below
Transfer time	Time to read/write the data once positioned	μs per sector

Worked example 3 — rotational latency. At 7200 RPM:

One revolution = $60 / 7200 \text{ s} = 8.33 \text{ ms}$.

Average rotational latency = half of that = 4.17 ms.

At 5400 RPM: $60/5400 = 11.1 \text{ ms} \rightarrow$ average **5.56 ms**.

At 15 000 RPM: $60/15000 = 4 \text{ ms} \rightarrow$ average **2 ms**.

Worked example 4 — full access time. Average seek 9 ms, 7200 RPM, transfer rate 100 MB/s, reading 4 KB.

Seek = 9 ms; latency = 4.17 ms; transfer = $4 \text{ KB} \div 100 \text{ MB/s} \approx 0.04 \text{ ms}$.

Total $\approx 13.2 \text{ ms}$. Note that **seek and latency dominate completely** — which is why sequential access is enormously faster than random access, and why disk scheduling algorithms exist.

Formatting and disk structure

Term	Meaning
------	---------

Low-level format	Writes the physical sector structure (headers, ECC). Done at the factory on modern drives
Partitioning	Divides the drive into logical volumes. The MBR (Master Boot Record) in sector 0 holds the boot code and the partition table (max 4 primary partitions); GPT is the modern replacement
High-level format	Creates the file system — FAT12/16/32, NTFS, ext4 — writing the boot sector, the allocation tables and the root directory
Boot sector	The first sector of a partition; holds the volume parameters and the bootstrap code
FAT / MFT	The map recording which clusters belong to which file
Bad sector	A sector that fails ECC; the drive remaps it to a spare
Defragmentation	Rearranging file clusters to be contiguous, reducing seek time. Useful on HDDs, harmful and unnecessary on SSDs

Solid State Drives (SSD) — for contrast

	HDD	SSD
Technology	Spinning magnetic platters + moving heads	NAND flash , no moving parts
Access time	5–15 ms	25–100 µs (~100× faster)
Random I/O	Poor (seek dominated)	Excellent
Shock resistance	Poor	Excellent
Noise / heat / power	Audible, warm, higher	Silent, cool, lower
Cost per GB	Much lower	Higher
Capacity per unit cost	Higher	Lower
Wear	Mechanical wear	Limited write cycles per cell — managed by wear levelling and TRIM
Defragmentation	Beneficial	Never — it just consumes write cycles

⚠ **2024 asked "which is *not* a property of an SSD"** with "less reliable" as the odd one out. SSDs are **more** reliable in the mechanical sense (no moving parts), though flash cells do wear.

8.8 Drive interfaces

Interface	Full name	Devices per channel	Connector	Speed	Notes

IDE / ATA / PATA	Integrated Drive Electronics / (Parallel) AT Attachment	2 — master and slave , set by jumpers or cable select	40-pin ribbon; 80-conductor cable for UDMA/66+	ATA-33 to UDMA/133 (up to 133 MB/s)	The controller moved onto the drive itself — hence "integrated". Two channels on a motherboard: primary and secondary , so 4 devices total
SATA	Serial ATA	1 per port (point-to-point)	7-pin thin data cable + 15-pin power	SATA I 1.5 Gb/s, II 3 Gb/s, III 6 Gb/s	Thinner cables, better airflow, hot-swappable , no jumpers. Superseded PATA entirely
SCSI	Small Computer System Interface	7 or 15 devices on one bus, each with a unique SCSI ID ; the bus must be terminated at both ends	50-pin (SCSI-1), 68-pin (Wide)	5 – 320 MB/s	Servers and high-end workstations; supports drives, scanners, tape
SAS	Serial Attached SCSI	Point-to-point, expanders allow many	Like SATA	3–24 Gb/s	Enterprise successor to SCSI
NVMe	Non-Volatile Memory express	Over PCIe	M.2 / U.2	Several GB/s	Modern SSD interface; bypasses the AHCI/SATA bottleneck
USB	Universal Serial Bus	127 via hubs	4-pin (USB 2.0)	1.5/12/480 Mb/s, USB 3.0 5 Gb/s	External drives; hot-pluggable; supplies 5 V power

PATA vs SATA is a guaranteed 5-marker. Answer with: parallel vs serial transmission, ribbon vs thin cable, 40 pins vs 7 pins, master/slave jumpers vs one device per port, 133 MB/s vs 600 MB/s, not hot-swappable vs hot-swappable.

⚠ **2024 trap:** a question defined "an electronic interface standard that defines the connection between a bus on a computer's motherboard and the computer's disk storage devices" and offered ATA / SATA / PATA / IDE. The generic, encompassing answer is **ATA**; IDE is its common informal name and PATA/SATA are its two forms. If forced, choose **ATA** for the generic definition. ⚠ verify against the official key if you can get it.

8.9 Optical storage — CD and DVD

Physical structure

A CD is a **1.2 mm polycarbonate disc**, **120 mm in diameter**, built up in layers:

```

label / screen printing
-----
lacquer protective coat
-----
reflective ALUMINIUM layer      <- the pits and lands are moulded here
-----
POLYCARBONATE substrate  1.2 mm

```

```

-----
^   laser reads from BELOW, through the plastic
|   (so a scratch on the LABEL side is worse than one on the read side --
|   the reflective layer is right under the label)

```

Data is stored on **one continuous spiral track** running **from the centre outwards** — unlike a hard disk's concentric circles. The spiral is about **5 km long** on a CD.

The read mechanism — how pits and lands become bits

1. A **laser diode** emits a beam; a collimating lens and beam splitter focus it through the polycarbonate onto the reflective layer.
2. The track consists of **pits** (depressions) and **lands** (the flat areas between them).
3. A pit is made **one quarter of the laser's wavelength deep**. Light reflecting off a pit therefore travels **half a wavelength further** (down and back up) than light off the surrounding land, so the two reflections are **180° out of phase and interfere destructively** — the reflected intensity drops sharply.
4. A **photodiode** measures the reflected intensity: strong = land, weak = pit.
5. **Crucially: a bit is not "pit = 1, land = 0"**. A **transition** from pit to land or land to pit represents a **1**; no transition represents a **0**. The encoding is **EFM (Eight-to-Fourteen Modulation)**, which guarantees a minimum and maximum run length so the drive can recover the clock.
6. The drive uses **CLV (Constant Linear Velocity)** for audio — spinning slower as the head moves outwards so the track passes at a constant speed — or CAV for data drives.

The write mechanisms

Media	Recording layer	How it is written	Rewritable?
CD-ROM	Aluminium, physically stamped from a glass master at manufacture	Not written by the user	No
CD-R	An organic dye layer (cyanine, phthalocyanine, azo)	A high-power laser burns/darkens the dye at chosen points, creating permanently low-reflectivity spots that mimic pits	No — write once (WORM)
CD-RW	A phase-change alloy (Ag-In-Sb-Te)	Three laser powers. Write: heat above the melting point (~600 °C) then cool fast → amorphous , non-reflective (a "pit"). Erase: heat to ~200 °C and cool slowly → crystalline , reflective (a "land"). Read: low power, no change	Yes , ~1000 cycles
DVD-RAM	Phase change, with embedded sector servo	Same principle, random access	Yes, ~100 000 cycles

CD vs DVD vs Blu-ray

The one sentence that explains the whole progression: *a shorter laser wavelength can be focused to a smaller spot, so pits can be made smaller and packed closer, so more data fits on the same-sized disc.*

Feature	CD	DVD	Blu-ray
Laser wavelength	780 nm (infra-red)	650 nm (red)	405 nm (blue-violet)
Numerical aperture	0.45	0.60	0.85
Track pitch	1.6 μm	0.74 μm	0.32 μm
Minimum pit length	0.83 μm	0.40 μm	0.15 μm
Capacity, single layer	650 – 700 MB	4.7 GB	25 GB
Dual layer	—	8.5 GB	50 GB
Double-sided dual-layer	—	17 GB	—
1× data rate	150 KB/s	1.38 MB/s	4.5 MB/s
Layer depth from surface	1.2 mm	0.6 mm	0.1 mm
Diameter / thickness	120 mm / 1.2 mm	120 mm / 1.2 mm (two 0.6 mm halves bonded)	120 mm / 1.2 mm

The 1× speed figures matter. A "52×" CD drive reads at $52 \times 150 \text{ KB/s} \approx \mathbf{7.8 \text{ MB/s}}$. A "16×" DVD drive reads at $16 \times 1.38 \approx \mathbf{22 \text{ MB/s}}$. Note that **1× is not the same speed for CD and DVD** — a common trap.

DVD variants: DVD-ROM, DVD-R and DVD+R (write once, two competing standards), DVD-RW and DVD+RW (rewritable), DVD-RAM (random access, cartridge). **DVD-5** = 4.7 GB single-sided single-layer; **DVD-9** = 8.5 GB single-sided dual-layer; **DVD-10** = 9.4 GB double-sided; **DVD-18** = 17 GB double-sided dual-layer.

⚠ **2024 asked a capacity-matching question** whose key gave **CD $\approx$ 640 MB, DVD 4.7 GB, floppy 1.4 MB and Blu-ray $\approx$ 4× a DVD**. Note that "CD = 640 MB" and "CD = 700 MB" are both used in textbooks (640/650 MB is the 74-minute disc, 700 MB the 80-minute one). Read the options and pick the closest offered.

8.10 Other storage

Device	Notes
Magnetic tape	Sequential access only ; very cheap per GB; huge capacity (LTO-9 = 18 TB native); used for archival backup . Access time in seconds or minutes
Pen drive / USB flash drive	NAND flash + USB controller; 4 GB – 2 TB; hot-pluggable
Memory cards	SD, microSD, CF; NAND flash; cameras and phones
Zip disk	100/250/750 MB removable magnetic cartridge; a 1990s floppy replacement; obsolete
Cloud storage	Remote storage accessed over a network; no local device

Likely exam questions

- Differentiate between SRAM and DRAM. [5]
- Differentiate between SDRAM and DDR RAM. [5]
- Explain the terms track, sector, cylinder and cluster. [5]
- Compare cache, primary and secondary memory with respect to capacity and speed. [5]
- Explain the ROM family: ROM, PROM, EPROM, EEPROM and Flash. [10]
- Describe the components of a hard disk and the organisation of data on it. [10]
- Explain the read and write mechanism of a CD and a DVD. [10]
- Explain the various storage devices, classifying them and comparing capacity, speed and cost. [15]

MCQ traps

- **EPROM = UV erase, whole chip. EEPROM = electrical erase, byte by byte.** The single "E" difference is the whole question.
- **DDR = two transfers per clock, not double the clock.**
- **Cluster ≠ sector.** The disk is formatted into sectors; the file system allocates clusters.
- **3.5" HD floppy = 1.44 MB.** Not 1.2 MB (that is the 5.25" HD).
- **CD 1× = 150 KB/s but DVD 1× = 1.38 MB/s.**
- **Blu-ray uses a 405 nm blue-violet laser** — shorter wavelength, higher capacity.
- **ROM is non-volatile; cache and all RAM are volatile.**
- **A hard disk is sealed but not evacuated** — it needs air for the heads to fly on.

9. Components of the motherboard

This is the highest-value section in the file. It is a near-certain 15-mark question, a reliable source of 8–12 MCQs, and the material you are least likely to already know. Read it twice and draw the diagram three times.

9.1 What the motherboard is, and the one idea behind its layout

The **motherboard** (also **mainboard**, **system board**, or **planar board**) is the main printed circuit board that **integrates all of the PC's primary system components on a single board** and provides the electrical pathways — the **buses** — over which they communicate.

The classic design has one organising principle, and if you understand it you can reconstruct the whole diagram from memory:

Fast things go near the CPU; slow things go far from it.

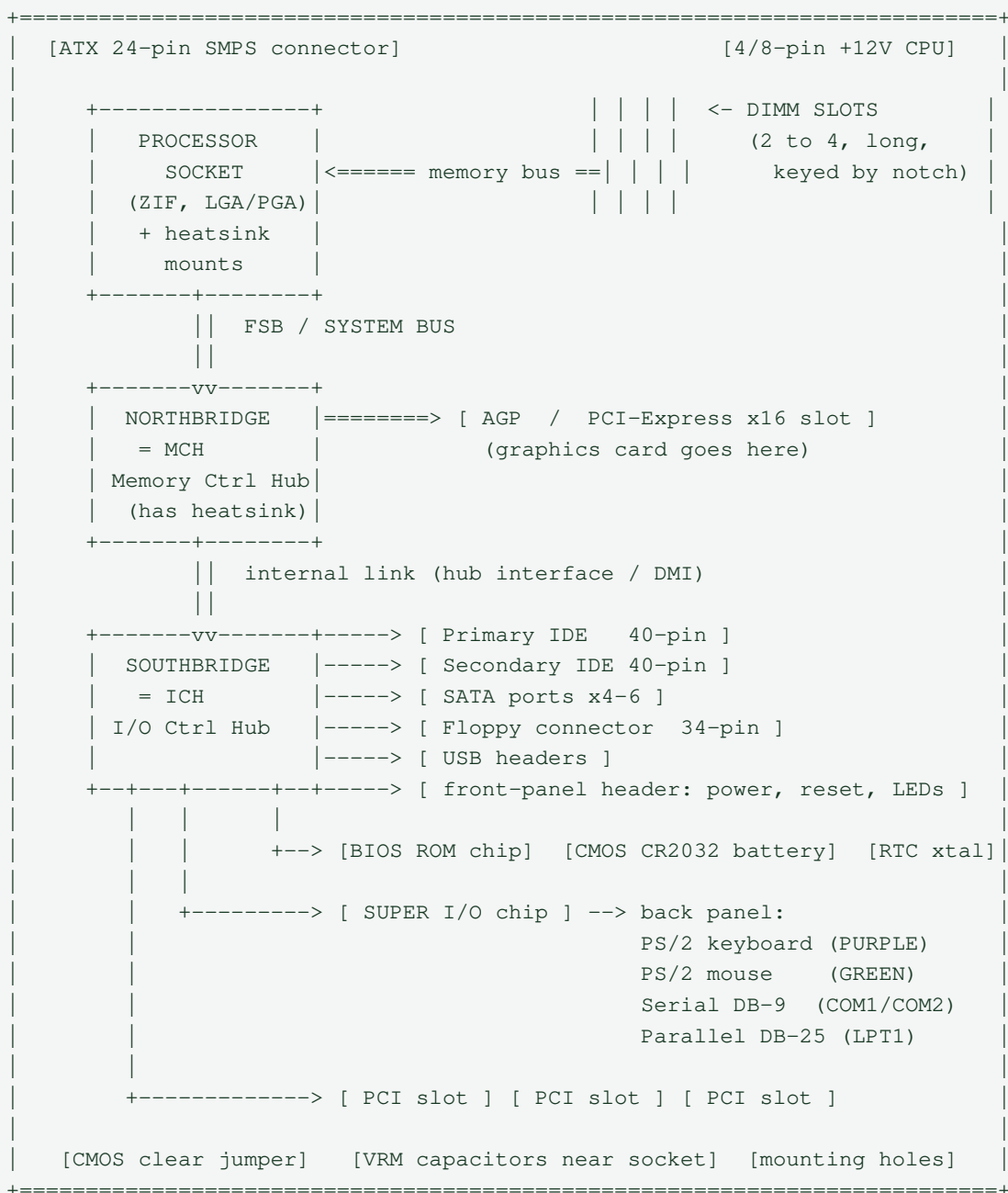
The chipset is therefore split into two halves:

- The **Northbridge**, or **Memory Controller Hub (MCH)**, handles the things that must be fast — the **CPU, the RAM and the graphics card**. It sits physically close to the CPU socket (which is why it is "north" on a diagram drawn with the CPU at the top) and usually has its own heatsink because it runs hot.
- The **Southbridge**, or **I/O Controller Hub (ICH)**, handles everything slower — **disks, PCI slots, USB, serial and parallel ports, keyboard, mouse, audio, BIOS**. It sits further down the board and connects to the Northbridge by an internal link.

Everything else on the board hangs off one of those two hubs.

Modern note (worth one closing line in an answer, no more): since about 2008 the memory controller and the graphics link have migrated **onto the CPU die itself**, leaving a single chipset chip called the **PCH (Platform Controller Hub)**. The **Diploma syllabus explicitly names the MCH and the ICH**, so answer the two-chip model as the main body of your answer and add the modern note at the end to show awareness.

9.2 The diagram — described precisely enough to hand-draw



How to draw it in the exam, in order:

1. Draw a large rectangle for the board. Add four small circles for **mounting holes**.

2. **Top-left:** a square for the **processor socket**, labelled "ZIF socket / LGA".
3. **Top-right of the socket:** 2–4 long thin parallel rectangles = **DIMM slots**. Note the little notch.
4. **Top edge:** a wide connector = **ATX 24-pin SMPS**; near the socket a small one = **4-pin +12 V**.
5. **Below the socket:** a square labelled **Northbridge / MCH**, with a small heatsink drawn on it. Join it to the socket with a thick double line labelled **FSB / system bus**, to the DIMMs with a line labelled **memory bus**, and to a slot on the right labelled **AGP / PCIe x16**.
6. **Below that:** a square labelled **Southbridge / ICH**, joined to the Northbridge.
7. **From the Southbridge**, draw lines to: **Primary IDE (40-pin)**, **Secondary IDE (40-pin)**, **SATA ports**, **floppy connector (34-pin)**, a row of **PCI slots** on the left, and the **Super I/O chip** which in turn feeds the back-panel **PS/2**, **serial and parallel** ports.
8. Add the three small items near the Southbridge: **BIOS ROM chip**, **CMOS battery (round)**, and the **RTC crystal**.
9. Label the whole thing and add a legend for the buses.

Label everything. In a hand-marked paper, a well-labelled diagram scores faster than three paragraphs of prose.

9.3 The Memory Controller Hub (Northbridge) and what hangs off it

(a) System bus / Front Side Bus (FSB)

Concept. A **bus** is a shared set of parallel wires connecting components. Every bus has three functional parts:

Sub-bus	Direction	Carries	Determines
Address bus	Unidirectional (CPU → memory/IO)	The address to be accessed	How much memory can be addressed — n address lines → 2^n locations
Data bus	Bidirectional	The actual data	Word size / how much data moves at once
Control bus	Bidirectional	Read/Write, clock, interrupt, reset, bus request/grant signals	Timing and coordination

Worked: a CPU with a **20-bit address bus** can address $2^{20} = 1 \text{ MB}$. With **32 bits**, $2^{32} = 4 \text{ GB}$ — which is exactly why 32-bit systems cannot use more than 4 GB of RAM. That one sentence is worth a mark in any bus question.

The **FSB (Front Side Bus)** is the bus between the CPU and the Northbridge; its speed (e.g. 533/800/1066/1333 MHz) was a headline specification of Pentium-era systems. Related buses to name: the **memory bus** (Northbridge ↔ RAM), the **backside bus** (CPU ↔ L2 cache), **expansion buses** (PCI, AGP, PCIe, ISA, USB, SCSI), and the **local bus** (a fast bus running at or near CPU speed, e.g. VESA local bus, PCI).

Bus bandwidth = bus width (bytes) × bus clock (MHz) × transfers per clock.

Worked: a 64-bit (8-byte) FSB at 200 MHz with quad pumping = $8 \times 200 \times 4 = 6400 \text{ MB/s}$ ("FSB 800").

(b) Processor socket

Aspect	Detail
Purpose	Holds the CPU and connects it electrically to the board
ZIF	Zero Insertion Force — a lever lifts to release all contacts so the chip drops in without pressing pins. Prevents bent pins
PGA	Pin Grid Array — pins on the processor , holes in the socket (AMD's approach)
LGA	Land Grid Array — flat pads on the processor , pins in the socket (Intel's modern approach)
Named examples	Socket 7, Socket 370, Socket A (462), Socket 478 , LGA 775 , LGA 1155, LGA 1200, LGA 1700, AM4, AM5
Around the socket	Heatsink/fan retention holes or bracket , the VRM (Voltage Regulator Module — capacitors and MOSFETs that step the 12 V supply down to the ~1 V the core needs), the CPU fan header, and thermal-sensor traces
Key rule	A motherboard supports exactly one socket type — CPU and board must match. One CPU per socket

(c) DIMM slots

Aspect	Detail
Name	Dual Inline Memory Module slot — the RAM slot
Why "dual inline"	Contacts on the two faces of the module are electrically independent (on the older SIMM they were tied together)
Pin counts	168 (SDRAM) · 184 (DDR) · 240 (DDR2 and DDR3, different notch positions) · 288 (DDR4/DDR5)
Keying	A notch in a generation-specific position physically prevents inserting the wrong module type
Retention	Plastic clips at each end snap over the module's side notches
Count	Typically 2 or 4 slots, often colour-coded in pairs
Dual channel	Populating matched modules in the same-coloured pair of slots lets the memory controller access two modules simultaneously, roughly doubling bandwidth
Laptop version	SO-DIMM — physically shorter

(d) AGP — Accelerated Graphics Port

Aspect	Detail
Purpose	A dedicated, point-to-point port for the graphics card only — created because 3-D graphics were saturating the shared PCI bus
Base spec	32-bit wide, 66 MHz → 266 MB/s at 1×
Speeds	1× = 266 MB/s · 2× = 533 MB/s · 4× = 1066 MB/s · 8× = 2133 MB/s

Successor	PCI Express — a serial, point-to-point interconnect of independent lanes (x1, x4, x8, x16), each lane 250 MB/s each way at PCIe 1.0, doubling with each generation. Not a bus at all despite the name
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Older bus for contrast: ISA (Industry Standard Architecture) — 8-bit/16-bit, 8 MHz, 8 MB/s, black slots, manual jumper configuration. Named in some syllabi; know the one line.

(c) Serial ports

Aspect	Detail
Also called	COM port , RS-232 port
Connector	DB-9 male on the PC's back panel (older equipment used DB-25)
Transmission	One bit at a time over a single data line, with start, data, parity and stop bits framing each character
Speed	300 bps to 115.2 kbps
Distance	Up to ~15 m — better than parallel, because fewer wires means less crosstalk
Names	COM1, COM2, COM3, COM4
Controlled by	The UART (Universal Asynchronous Receiver/Transmitter) inside the Super I/O chip
Devices	Mouse (legacy), external modem, serial printer, UPS, network/router console, industrial instruments, GPS

(d) Parallel port

Aspect	Detail
Also called	LPT port , Centronics port, printer port
Connector	DB-25 female on the PC (the printer end is a 36-pin Centronics)
Transmission	8 bits (one byte) simultaneously over 8 parallel data lines
Speed	150 KB/s (SPP) up to 2 MB/s (ECP)
Distance	Limited to ~3–5 m — crosstalk and skew between the parallel wires
Modes	SPP (Standard, unidirectional), EPP (Enhanced, bidirectional, for non-printers), ECP (Extended Capability, bidirectional with DMA)
Names	LPT1, LPT2
Devices	Printers, scanners, Zip drives, dongles

Serial vs parallel — the classic comparison, and the counter-intuitive punchline.		
	Serial	Parallel
Bits at a time	1	8

Wires	Few	Many
Connector	DB-9	DB-25
Max cable length	~15 m	~3–5 m
Speed	Slower per clock	Faster per clock
Cost	Cheaper	Costlier

Parallel sounds obviously faster — so why is every modern high-speed interface (SATA, USB, PCIe, HDMI) serial? Because at high frequencies the bits on parallel wires arrive at slightly different times (**skew**) and interfere with each other (**crosstalk**), and both problems get worse with speed and length. A single serial line can be clocked far higher than a parallel bundle can be kept in step. **That is the insight the examiner is looking for if the question asks "why has serial replaced parallel".**

(e) PS/2 ports

Aspect	Detail
Origin	IBM Personal System/2 , 1987
Connector	6-pin mini-DIN , round
Colour code	PURPLE = keyboard · GREEN = mouse — memorise this, it is asked
Signals	Clock, Data, +5 V, Ground, and two unused pins
Protocol	Bidirectional synchronous serial
Interrupts	Keyboard IRQ 1 , mouse IRQ 12
Not hot-pluggable	Connecting one with the PC on can fail to detect it or damage the port — plug in before booting
Status	Superseded by USB, but persists on some boards (and is preferred by some gamers for its interrupt-driven, N-key-rollover behaviour)

(f) USB — Universal Serial Bus

Aspect	Detail
Purpose	One standard connector to replace serial, parallel, PS/2 and game ports
Topology	Tiered star , with hubs. Up to 127 devices , maximum 5 tiers
Connector	4 pins for USB 1.1/2.0: VBUS (+5 V) , D– , D+ , GND . USB 3.0 adds 5 more (9 total)
Speeds	USB 1.1 Low Speed 1.5 Mbps / Full Speed 12 Mbps · USB 2.0 High Speed 480 Mbps · USB 3.0 SuperSpeed 5 Gbps · USB 3.1 10 Gbps · USB 3.2 20 Gbps · USB4 40 Gbps
Power	Supplies 5 V , 500 mA (USB 2.0) / 900 mA (USB 3.0); USB-C Power Delivery up to 100 W+

⚠ **Beep codes differ between AMI, Award and Phoenix BIOSes.** If a question asks for an exact code, answer with the pattern above and note the vendor dependence. The one you can rely on is **one short beep = all is well**.

The BIOS boot sequence — get the order right

The examiner's favourite ordering question. The sequence is:

1. **Power on** → SMPS stabilises and asserts the **Power Good** signal.
2. **CPU reset** — the processor jumps to the BIOS entry point at address `FFFF:0000`.
3. **BIOS start-up / initialisation**.
4. **POST** — the self test.
5. **System check** — detect and initialise devices; display the BIOS screen.
6. Read the **boot order** from CMOS.
7. Load the **boot sector (512 bytes)** of the first bootable device to memory at `0000:7C00`.
8. Transfer control to the bootstrap loader, which loads the operating system.
9. The OS takes over.

9.6 CMOS battery

Aspect	Detail
What CMOS is	A small block of volatile CMOS SRAM (traditionally 64 or 128 bytes) inside the Southbridge/RTC chip, holding the BIOS settings
Why a battery	The settings and the clock must survive with the PSU switched off, so they need standby power
The cell	CR2032, 3 V lithium coin cell — the number encodes 20 mm diameter, 3.2 mm thick
Life	3–10 years
Symptoms of failure	Date and time reset at every power-off · "CMOS checksum error" / "CMOS battery low" at boot · BIOS settings revert to defaults (boot order, drive settings) · the machine may refuse to boot from the right device. It does NOT erase the hard disk or the RAM
CMOS clear	A 3-pin jumper near the battery, or removing the cell for a few minutes, resets the BIOS to defaults — the standard way to clear a forgotten BIOS password
Key distinction	BIOS = the program, in non-volatile ROM. CMOS = the settings, in volatile RAM kept alive by the battery. This distinction is very commonly examined

9.7 Real Time Clock (RTC)

Aspect	Detail
Function	Keeps the date and time running continuously , independent of whether the computer is on or which OS is installed

Implementation	A counter chip integrated into the Southbridge (historically the Motorola MC146818), sharing the CMOS RAM and the battery
Crystal	Driven by a 32.768 kHz quartz crystal — chosen because $32,768 = 2^{15}$, so a simple 15-stage binary divider produces an exact 1 Hz pulse. Learn this number and the reason; it makes a neat exam point
Powered by	The CMOS battery when the PSU is off
Provides	Time of day, calendar date, and a periodic interrupt (IRQ 8) usable for alarms and OS timing
Related but different	The system clock generator — a separate, much higher-frequency crystal (typically 14.318 MHz) whose output is multiplied by a PLL to produce the CPU, FSB and bus clocks. Do not confuse the RTC (timekeeping) with the system clock (timing/synchronisation) . A frequent trap

9.8 SMPS connector

Aspect	Detail
SMPS	Switched Mode Power Supply — converts 230 V AC mains into the low-voltage DC rails the PC needs, by rectifying, then switching at high frequency through a small transformer, then rectifying and filtering again. High-frequency switching is why the transformer can be small and the efficiency high (~80–90%)
Functions of a PSU	Rectification (AC → DC) · Filtering (smoothing ripple) · Voltage regulation (holding rails steady under changing load) · Transformation/stepping down · Protection (over-voltage, over-current, short circuit). Cooling is done by its fan but is not a power-conversion function — 2024 used "cooling" as the odd one out
Main connector	ATX 20-pin , extended to 24-pin in ATX12V 2.0 (the extra 4 pins add more +12 V, +5 V and +3.3 V for PCIe). Keyed and clipped so it cannot go in backwards
CPU connector	4-pin (ATX12V) or 8-pin (EPS12V) +12 V connector near the CPU socket, feeding the VRM
Other connectors	PCIe 6-pin / 8-pin for graphics cards · SATA power 15-pin flat · Molex 4-pin for older drives and fans · 4-pin Berg for floppy drives
Rails	+3.3 V (chipset, RAM, some logic) · +5 V (logic, USB, older drives) · +12 V (CPU VRM, motors, fans, graphics) · –12 V (legacy serial ports) · +5 VSB (5 V standby) — always live while the mains is connected; powers Wake-on-LAN and the power button)
PS_ON	A green wire pulled low by the power button to switch the supply on — this is why an ATX machine can be switched off by software ("It is now safe to turn off your computer" belonged to the older AT standard, which could not)
Power Good (PWR_OK)	A grey signal the PSU asserts once all rails are stable; the chipset holds the CPU in reset until it appears. Without it the machine will not start
Form factors	AT (obsolete, split P8/P9 connectors) · ATX · micro-ATX · SFX · TFX

MCQ traps

- **Northbridge = MCH = memory, CPU, graphics. Southbridge = ICH = disks, PCI, USB, ports.** If the question mentions *cache, main memory or the PCI bus controller*, the answer is **north bridge**.
- **Floppy = 34 pins, IDE = 40 pins, SCSI-1 = 50 pins, ATX main = 20/24 pins.** Four numbers.
- **80-conductor cable, 40-pin connector.** Both numbers are correct simultaneously.
- **PS/2: purple keyboard, green mouse.**
- **CMOS battery failure loses the *settings and clock*, not your data.**
- **BIOS ≠ CMOS.** Program vs settings.
- **RTC crystal = 32.768 kHz**, not the system clock frequency.
- **The SMPS is not a motherboard component** — only its connector is.
- **A serial cable can be longer than a parallel one**, despite parallel being "faster".

10. Numbers you must memorise

Cover the right column. Recite. Repeat daily from 20 September.

Data representation

Fact	Value
Bit / nibble / byte	1 / 4 / 8 bits
Word length	16, 32 or 64 bits (machine dependent)
1 KB / MB / GB / TB	2^{10} / 2^{20} / 2^{30} / 2^{40} bytes
ASCII bits / characters	7 bits / 128 (extended 8 bits / 256)
EBCDIC	8-bit, IBM
Unicode UTF-8 / UTF-16 / UTF-32	1–4 bytes / 16 bits (BMP) / 32 bits
BCD valid codes	0000–1001 only ; 1010–1111 invalid
ASCII A / a / 0 / space	65 / 97 / 48 / 32
8-bit 2's complement range	–128 to +127
8-bit 1's complement range	–127 to +127

Generations

Fact	Value
1st / 2nd / 3rd / 4th generation technology	Vacuum tube / transistor / IC / VLSI microprocessor
Transistor invented	Bell Labs, 1947

USB 1.1 / 2.0 / 3.0 speeds	1.5 & 12 Mbps / 480 Mbps / 5 Gbps
USB supply voltage	5 V
PCI width / clock / bandwidth	32-bit / 33 MHz / 133 MB/s
AGP 1× / 2× / 4× / 8×	266 / 533 / 1066 / 2133 MB/s
DIMM pins SDR / DDR / DDR2-3 / DDR4	168 / 184 / 240 / 288
ATX main power connector	20-pin (24-pin in ATX12V 2.0)
CPU auxiliary power connector	4-pin or 8-pin +12 V
PSU rails	+3.3, +5, +12, –12, +5 VSB
CMOS battery	CR2032, 3 V lithium
RTC crystal	32.768 kHz (= 2 ¹⁵ Hz)
System clock crystal	14.318 MHz
Normal POST result	One short beep
Continuous short beeps	Power supply / system board failure
BIOS entry vector	FFFF:0000
Boot sector load address / size	0000:7C00 / 512 bytes
ATX board size	305 × 244 mm

Devices

Fact	Value
Keyboard keys	101 / 104 (older 83/84)
Function keys	12 (F1–F12)
Numeric keypad keys	17
DMP pins	9 or 24
DMP / inkjet / laser speed units	CPS / PPM / PPM
Line printer speed unit	LPM
Laser fusing temperature	~200 °C
Printer resolution unit	DPI

MICR font

E-13B

Where the rest of Paper-I lives

Paper-I unit	Covered in
§1 Introduction to computers	This file, §1–§5
§2 Input and output devices	This file, §6–§7
§3 Storage devices	This file, §8 · deeper: <code>CORE_02_ComputerOrganization.md</code>
§4 Components of the motherboard	This file, §9
§5 Computer networks	<code>CORE_05_ComputerNetworks.md</code> — and note that 2024 made this ~25% of Paper-I. Do not skim it
§6 Internet technology	<code>CSDip_03_SystemAnalysisAndViruses.md</code> §3
§7 Computer languages	Brief treatment in <code>CSDip_03</code> and <code>CORE_09_Programming_C_CPP.md</code>
§8 Computer software / MS Office	<code>CSDip_02_MSOoffice_and_DOS.md</code>
§9 Computer viruses	<code>CSDip_03_SystemAnalysisAndViruses.md</code> §1
Number systems, deeper	<code>CORE_01_DigitalLogic.md</code>
All drill questions	<code>CSDip_04_MCQ_and_QuestionBank.md</code>

A note on the file index. `CSDip_00_OVERVIEW.md` §9 lists a seven-file plan (`CSDip_03_Networks_and_Internet`, `CSDip_04_Languages_and_Software`, `CSDip_05_Viruses`, `CSDip_06_SystemAnalysisDesign`). The folder has since been consolidated into **five** files: the viruses, System-Analysis-and-Design and Internet-technology material is combined in `CSDip_03_SystemAnalysisAndViruses.md`, and `CSDip_04` is now the **MCQ and question bank**. Networks are covered by `CORE_05` with the Diploma-specific angles noted in `CSDip_03`. **All the syllabus content is still covered — only the filenames changed.**